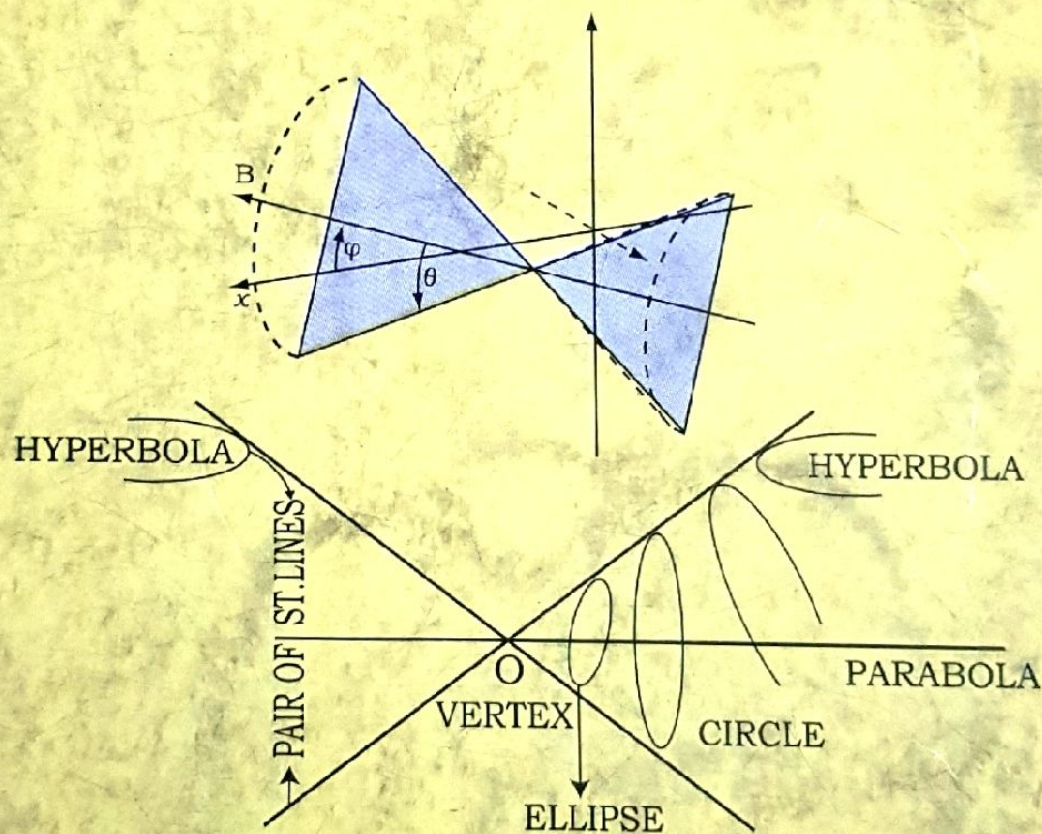


A TEXT BOOK ON CO-ORDINATE GEOMETRY (TWO AND THREE DIMENSIONS) WITH VECTOR ANALYSIS



RAHMAN & BHATTACHARJEE

Art. 31. Transformation of Co-ordinates.

The co-ordinates of a point or the equation of a curve are always given with reference to a fixed origin and a set of axes of co-ordinates. The above co-ordinates of the equation of the curve changes when the origin is changed or the direction of axes changed or both. The process of changing the co-ordinate of a point or the equation of a curve is called **transformation of co-ordinates**. Now we have to investigate the mode of the change of the co-ordinates or the equation of the curve according to the transfer from one set to another set of co-ordinate axes.

Art. 32. Change of origin (Translation of axes)

To find the change in the co-ordinates of a point when the origin is shifted to another point $O'(\alpha, \beta)$ where the direction of axes remains unaltered.

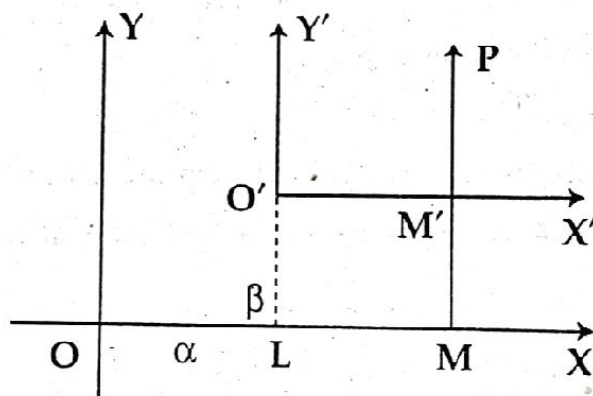


Fig. 10

Let us take a new pair of axes $O'X'$ and $O'Y'$ parallel to the old pair OX and OY ; O' being the new origin whose co-ordinates are (α, β) referred to $O'X'$ and $O'Y'$.

Let (x', y') be the co-ordinates referred to the axes $O'X'$ and $O'Y'$ of a point P , whose co-ordinates referred to the old axes are (x, y) .

It is required to transform the co-ordinates (x, y) in terms of (x', y') .

From O' and P draw $O'L$ and PM perpendiculars to OX . Let PM meet $O'X'$ in M' .

Then $OL = \alpha, LO' = \beta, OM = x, MP = y$

Also $O'M' = x'$ and $M'P = y'$

Therefore, $OM = OL + LM = OL + O'M'$

$$\therefore x = \alpha + x'$$

Similarly, $MP = MM' + M'P = LO' + M'P$

$$\therefore y = \beta + y'$$

The transformed co-ordinates are

(i)

(ii)

$$\text{and } \left. \begin{aligned} x' &= x - \alpha \\ y' &= y - \beta \end{aligned} \right\} \dots \dots \dots (2)$$

Rule : In order to shift the origin to (α, β) the transformation is obtained by replacing x by $x + \alpha$ and y by $y + \beta$. If from the transformed equation we want to get the old equations then replace x by $x - \alpha$ and y by $y - \beta$. This is known as shifting the origin back.

Art. 33. Rotation of axes (origin fixed)

To find the change in the co-ordinates of a point when the direction of axes is turned through an angle θ where as the origin of co-ordinates remains the same.

Let OX and OY be the old axes and OX' and OY' set the new axes. O is the common origin for the two sets of axes. Let the angle $X'OX$ through which the axes have rotated be represented by θ .

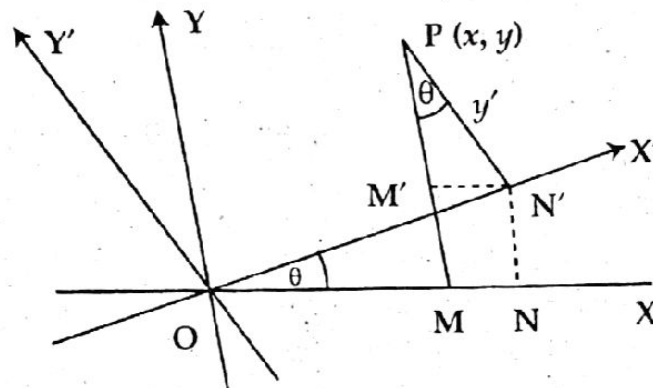


Fig. 11

Let P be any point in the plane and let its co-ordinates referred to the old axes be (x, y) , and referred to the new axes be (x', y') .

Let us try to determine x and y in terms of x', y' and θ . Draw PM perpendicular to OX , PN' perpendicular to OX' , and $N'N$ perpendicular to OX , and $N'M'$ parallel to OX .

$$\begin{aligned} \text{Then } x &= OM = ON - MN = ON - M'N' \\ &= ON' \cos \theta - PN' \sin \theta = x' \cos \theta - y' \sin \theta \end{aligned}$$

$$\begin{aligned} y = MP &= MM' + M'P = NN' + M'P \\ &= ON' \sin \theta + PN' \cos \theta = x' \sin \theta + y' \cos \theta \end{aligned}$$

Hence the formula for the rotation of the axes through an angle θ are.

$$\text{and } \left. \begin{aligned} x &= x' \cos \theta - y' \sin \theta \\ y &= x' \sin \theta + y' \cos \theta \end{aligned} \right\} \dots \dots \dots (3)$$

Art. 34. Change of origin with the change of the direction of axes.

It is really the combination of Art. 32 and Art. 33. The best method is to apply Art. 32 first and then Art. 33. Of course the two transformations may also be made simultaneously.

Let us suppose that the system of axes be rectangular. The origin is shifted to the point (α, β) and then the axes are rotated through an angle θ . If the co-ordinates of any point be (x, y) in the old system, and (x', y') in the new system, from Art. 32, and Art. 33.

we have,

$$\text{and } \begin{cases} x = \alpha + x' \cos \theta - y' \sin \theta \\ y = \beta + x' \sin \theta + y' \cos \theta \end{cases} \quad \dots \quad (4)$$

Q2. Art. 35. Invariants.

[R.U. 1983]

If by the rotation of the rectangular co-ordinate axes about the origin, the expression $ax^2 + 2hxy + by^2$ changes to

$$a'x'^2 + 2h'x'y' + b'y'^2, \quad a + b = a' + b' \text{ and } ab - h^2 = a'b' - h'^2$$

Soln. Let (x, y) be the co-ordinates of a point P referred to a set of rectangular axes. If the axes are rotated through an angle θ about the origin, let the co-ordinates of the same point P be (x', y') referred to the new system of rectangular axes. Then we have by

$$(3) \text{ Art. 33. } x = x' \cos \theta - y' \sin \theta \text{ and } y = x' \sin \theta + y' \cos \theta.$$

By using the above transformations the expression

$ax^2 + 2hxy + by^2$ becomes

$$\begin{aligned} & a(x' \cos \theta - y' \sin \theta)^2 + 2h(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) + b(x' \sin \theta + y' \cos \theta)^2 \\ &= a'x'^2 + 2h'x'y' + b'y'^2 \end{aligned}$$

$$\text{where } a' = a \cos^2 \theta + 2h \sin \theta \cos \theta + b \sin^2 \theta \quad \dots \quad (1)$$

$$b' = a \sin^2 \theta - 2h \sin \theta \cos \theta + b \cos^2 \theta \quad \dots \quad (2)$$

$$h' = h(\cos^2 \theta - \sin^2 \theta) - (a - b) \sin \theta \cos \theta \quad \dots \quad (3)$$

From (1) and (2) we have on simplification

$$a' + b' = a + b$$

$$\text{Again } 2a' = 2a \cos^2 \theta + 4h \sin \theta \cos \theta + 2b \sin^2 \theta$$

$$= a(1 + \cos 2\theta) + 2h \sin 2\theta + b(1 - \cos 2\theta)$$

$$= a + b + 2h \sin 2\theta + (a - b) \cos 2\theta \quad \dots \quad (4)$$

$$\text{Similarly, } 2b' = a + b - 2h \sin 2\theta - (a - b) \cos 2\theta \quad \dots \quad (5)$$

From (4), (5) and (3) we have

$$4(a'b' - h'^2) = (a + b)^2 - \{2h \sin 2\theta + (a - b) \cos 2\theta\}^2$$

$$- \{2h \cos 2\theta - (a - b) \sin 2\theta\}^2$$

$$= (a + b)^2 - 4h^2 - (a - b)^2 = 4ab - 4h^2$$

$$\text{or, } a'b' - h'^2 = ab - h^2$$

The two quantities $a + b$ and $ab - h^2$ for the expression $ax^2 + 2hxy + by^2$ are called **invariants** of transformation from one system of rectangular axes (Without change of the origin), because their values remain unchanged by the transformation.

Q3. Cor. Removal of the-xy-term from the expression $ax^2 + 2hxy + by^2$

If we rotate the rectangular axes through an angle θ about the origin, we have transformations by Eq. 3, Art. 33.

$$x = x' \cos \theta - y' \sin \theta, \quad y = x' \sin \theta + y' \cos \theta$$

Putting the values of x and y in $ax^2 + 2hxy + by^2$

$$\text{we have, } a'x'^2 + 2h'x'y' + b'y'^2$$

where the values of a' , b' and h' are available from (1), (2) and (3) of Art 35.

The co-efficient of $x'y'$ term in the new expression is $2h'$ where $h' = h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta$. In order to remove the $x'y'$ term, the co-efficient h' is zero i.e.

$$h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta = 0$$

$$\therefore h\cos 2\theta = \frac{1}{2}(a-b)\sin 2\theta \text{ or, } \tan 2\theta = \frac{2h}{a-b}$$

$$\therefore \theta = \frac{1}{2}\tan^{-1} \frac{2h}{a-b} \quad \dots \quad \dots \quad \dots \quad (6)$$

Hence if the axes are rotated through an angle $\theta = \frac{1}{2}\tan^{-1} \frac{2h}{a-b}$ then xy term in the expression $ax^2 + 2hxy + by^2$ vanishes.

Ex. 1. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin is transferred to the point $(2, -1)$.

when the origin is transferred to the point $(2, -1)$ Put $x = x' + 2$, $y = y' - 1$ in the above equation. Then $2(x' + 2)^2 + 3(y' - 1)^2 - 8(x' + 2) + 6(y' - 1) - 7 = 0$

$$\text{or, } 2x'^2 + 3y'^2 = 18$$

Now removing suffixes the equation referred to new axes is $2x^2 + 3y^2 = 18$

Ex. 2. Determine the equation of the parabola

$$x^2 - 2xy + y^2 + 2x - 4y + 3 = 0 \quad \dots \quad \dots \quad \dots \quad (i)$$

after rotating of axes through 45°

When the axes have been rotated through an angle 45° , then

$$x = x' \cos 45^\circ - y' \sin 45^\circ = (x' - y')/\sqrt{2}$$

$$y = x' \sin 45^\circ + y' \cos 45^\circ = (x' + y')/\sqrt{2}$$

$$\text{Put them in (1), then } \left(\frac{x' - y'}{\sqrt{2}}\right)^2 - 2\left(\frac{x' - y'}{\sqrt{2}}\right)\frac{x' + y'}{\sqrt{2}} + \left(\frac{x' + y'}{\sqrt{2}}\right)^2 + 2\left(\frac{x' - y'}{\sqrt{2}}\right) - 4\left(\frac{x' + y'}{\sqrt{2}}\right) + 3 = 0$$

On simplification the equation reduces to

$$2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0$$

Now dropping the suffixes, the equation is

$$2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0 \text{ Ans.}$$

Ex. 3. Remove the first degree terms in

$$3x^2 + 4y^2 - 12x + 4y + 13 = 0$$

Complete the square,

$$\text{or, } 3(x^2 - 4x) + 4\left(y^2 + y + \frac{1}{4}\right) = 0$$

$$\text{or, } 3(x - 2)^2 + 4\left(y + \frac{1}{2}\right)^2 = 0 \quad \dots \quad \dots \quad \dots \quad (ii)$$

GENERAL EQUATION OF SECOND DEGREE

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Art. 43. Conic Section regarded as locus.

Definition : If a point P moves in a plane such a way that the ratio of its distance PS from a fixed point S in the plane to its perpendicular distance PM from a fixed straight line XM in it, is always a constant, the locus of the point P is called a Conic Section or briefly a Conic.

The constant ratio is called the **eccentricity** of the conic and is generally represented by the letter ' e '. The fixed point S is called the **focus** and the fixed straight line XM is called **directrix** of the conic.

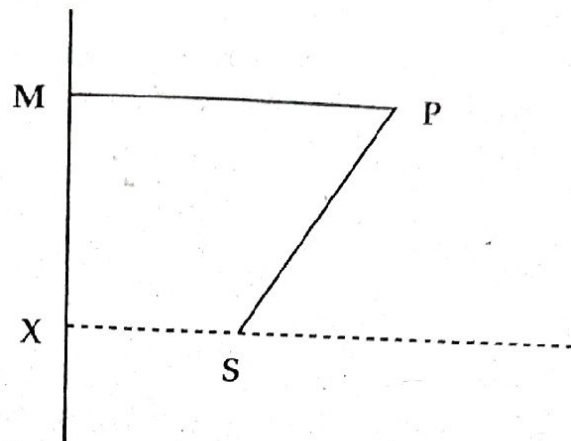


Fig. 16

The conic is called a **Parabola**, and **Ellipse** or a **Hyperbola** according as the eccentricity $e = , < \text{or} > .1$.

That is for a **Parabola**, $SP = PM$

for an **Ellipse**, $SP = ePM \quad e < 1$

for **Hyperbola**, $SP = ePM \quad e > 1$

The eccentricity is a positive number.

Art. 44. Conic section regarded as a section of a double right circular Cone.

Literally the term conic section means the section of a cone but the term is used in a restricted sense to mean the section of a particular kind of cone known as **Circular cone**. This is a cone generated by the revolution of an infinite straight line about a fixed point in such a way that it is always inclined at a constant angle to a fixed straight line passing through the fixed point.

The fixed point is called the **Vertex**, the fixed straight line is called the **Axis**, the revolving line is called **Generator** and the constant angle of inclination of the revolving line to the fixed line is called the **Semivertical Angle** of the Cone.

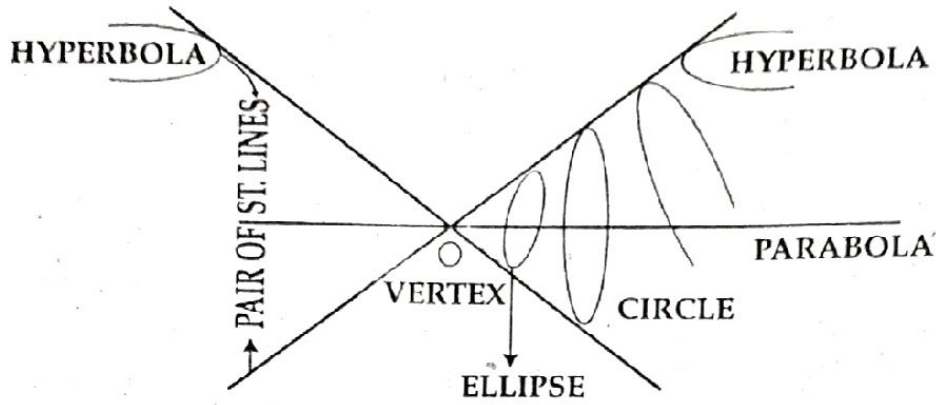


Fig: 17

If the cutting plane while passing through the vertex cuts the cone again, the curve section is a pair of straight lines intersecting at the vertex.

If the cutting plane without passing through the vertex is perpendicular to the axis the curve of the section is a Circle.

If the cutting plane neither passes through the vertex nor is perpendicular to the axis, the curve of the section is a parabola; an ellipse or a hyperbola according as the inclination of the cutting plane to the axis is equal, greater or less than the inclination of the generator to the same line.

The different positions of the cutting plane and the curves of sections obtained are shown in the above fig.

Let us explain the conics above in more details. The above figure is a double right circular cone and a plane cuts it in different positions giving different conic sections such as —

- (1) A pair of straight lines as obtained from the double cone by a section through the common vertex cutting the double cone in two generators.
- (2) A circle is obtained by a plane section perpendicular to the common axis.
- (3) A parabola is obtained by a plane section parallel to a generator of the double cone.
- (4) An ellipse is obtained by a section cutting only one half of the double cone, and not perpendicular to the axis or parallel to the generator.
- (5) A hyperbola is obtained by a section cutting both halves of the double cone and not passing through the common vertex.

Note : The circle is a degenerate form of ellipse and a pair of straight lines is a degenerate form of hyperbola.

Art. 44. (a) একটি সরল বৃত্তীয় কোণকে (Right Circular cone) এর কোন সমতল (Plane) দ্বারা বিভিন্ন ছেদিতাংশকে কোণকচ্ছেদ (Conic Section) বা সংক্ষেপে কণিক বলা হয়।

একটি সরলরেখা একটি নির্দিষ্ট বিন্দু দিয়ে এবং একটি নির্দিষ্ট সরলরেখার সাথে একটি ধ্রুব কোণে ঘুরে যে কোণক উৎপন্ন করে একে সরল বৃত্তীয় কোণক বলা হয়। নির্দিষ্ট বিন্দুটিকে বলা হয় শীর্ষক বিন্দু (Vertex), নির্দিষ্ট সরল রেখাকে বলা হয় অক্ষরেখা (Axis)। ঘূর্ণায়মান সরলরেখাকে বলা হয় কারিকা রেখা (Generating line) এবং অক্ষরেখার সাথে কারিকারেখার ধ্রুব কোণকে বলা হয় অর্ধশীর্ষ কোণ (Semivertical angle)।

Case II. Let either A or B be zero ; then eq. (3) takes the form if $A = 0$
or, $By^2 + 2Gx + 2Fy + C = 0$

or, $B(y + F/B)^2 = -2Gx - C + F^2/B = -2G\left(x + \frac{BC - F^2}{BG}\right) \dots \dots \dots (7)$

(i) If $G = 0$, then eq. (7) will represent a pair of parallel straight lines.

(ii) If $G = 0$ and $F^2/B - C = 0$, (7) will represent a pair of coincident lines.

(iii) $G \neq 0$, shift the origin to $\left(\frac{BC - F^2}{BG}, -\frac{F}{B}\right)$;

then eq. (7) takes the form $y^2 = -\frac{2G}{4}x \dots \dots \dots (8)$

which represents a parabola.

Since $AB = ab - h^2$ from (4)

or, $ab - h^2 = 0$, as A is zero

Hence when $ab - h^2 = 0$, conic (1) will represent a parabola.

Case III. When $A = B$ in eq. (3) i. e. when $a = b$ and $h = 0$ from (1), the eq. (1) will represent a circle.

Hence Eq. (1) represents an ellipse, parabola or hyperbola according as $ab - h^2 >, =$ or < 0

Art. 48. From the above discussions the general equation of the second degree.

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent

(1) a pair of straight lines if the determinant

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

(i) two parallel lines if $\Delta = 0, ab = h^2$

(ii) two perpendicular lines if $\Delta = 0, a + b = 0$

(2) a circle if $a = b, h = 0$,

(3) a parabola if $ab = h^2, \Delta \neq 0$

(4) an ellipse if $ab - h^2 > 0, \Delta \neq 0$

(5) a hyperbola if $ab - h^2 < 0, \Delta \neq 0$

(6) a rectangular hyperbola if $a + b = 0, ab - h^2 < 0, \Delta \neq 0$

Note : $\Delta = abc + 2fgh - af^2 - bg^2 - cg^2 =$

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Art. 49. Centre of a conic.

Definition : The centre of a conic is the point in the plane of the conic such that every chord of the conic passing through it is bisected at it.

Let the equation to the conic be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \dots \quad \dots \quad \dots \quad (2)$$

$$\text{and let } \frac{x - x_1}{l} = \frac{y - y_1}{m} = r \quad \dots \quad \dots \quad \dots \quad (3)$$

be the equation of any line through (x_1, y_1)

If the line (2) meets the conic (1) at a distance r from (x_1, y_1) then $(x = x_1 + lr, y = y_1 + mr)$ is a point of intersection of (1) and (2). The from (1)

$$a(x_1 + lr)^2 + 2h(x_1 + lr)(y_1 + mr) + b(y_1 + mr)^2 + 2g(x_1 + lr) + 2f(y_1 + mr) + c = 0$$

$$\text{or, } (al^2 + 2hlm + bm^2)r^2 + 2r\{ax_1 + hy_1 + g\}l + \{hx_1 + by_1 + f\}m + (ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c) = 0 \quad \dots \quad \dots \quad \dots \quad (4)$$

From Eq. (4) we see that r has two roots say r_1 and r_2 which are the distances from (x_1, y_1) i.e. the line (2) intersects the conic (1) at two points.

If (x_1, y_1) be the middle point of the chord, then r_1 and r_2 have equal but opposite sign i.e. $r_1 = -r_2$, or $r_1 + r_2 = 0$

The sum of the roots of r in (3) = 0

$$\text{or } (ax_1 + by_1 + g)l + (hx_1 + by_1 + f)m = 0 \quad \dots \quad \dots \quad \dots \quad (5)$$

If (x_1, y_1) is the centre of the conic, then the relation (4) holds good for all values of l and m

$$\therefore ax_1 + hy_1 + g = 0 \quad \dots \quad \dots \quad \dots \quad (6)$$

$$hx_1 + by_1 + f = 0 \quad \dots \quad \dots \quad \dots \quad (7)$$

The above two equations given the centre of the conic (1). Solve (6) and (7) for x_1 and y_1

$$\therefore x_1 = \frac{hf - bg}{ab - h^2}, y_1 = \frac{gh - af}{ab - h^2}$$

Hence the co-ordinates of centre (1) is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$

Cor. 1. When $ab - h^2 = 0$, the co-ordinates of the centre are at infinity and the curve is a parabola which has no centre.

Cor. 2. If $hf - bg = 0$ and $ab - h^2 = 0$

i.e. $a/h = h/b = g/f$

The equations (3) and (4) represent the same line. In this case any point on the straight line (3) or (4) may be regarded as the centre, for every such point satisfies the definition of the centre.

The locus in this case is a pair of parallel straight lines.

Note : The equations (3) and (4) can be obtained by, differentiating (1) partially with respect to x and y respectively.

$$\text{If } F(x, y) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{Then, } \frac{\partial F}{\partial x} \equiv 2(ax + hy + g) = 0, \quad \frac{\partial F}{\partial y} \equiv 2(hx + by + f) = 0,$$

Solve $\frac{\partial F}{\partial x} = 0$ and $\frac{\partial F}{\partial y} = 0$ for x and y . Thus the co-ordinates for the centre (x, y) are obtained.

Art. 50. Equation referred to centre of the conic. (Rectangular axes)

Let the equation to the conic be

$$F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \dots \dots (1)$$

$$\frac{\partial F}{\partial x} = 2(ax + hy + g) = 0, \quad \frac{\partial F}{\partial y} = 2(hx + by + f) = 0 \quad \dots \dots (2)$$

If (x_1, y_1) be the co-ordinates of the centre of the conic, then these co-ordinates are obtained by solving these equations,

$$ax_1 + hy_1 + g = 0 \text{ and } hx_1 + by_1 + f = 0 \quad \dots \dots (3)$$

If the origin is transferred to (x_1, y_1) the Eq. (1) becomes $ax^2 + 2hxy + by^2 + c_1 = 0$... (4)

where $c_1 = ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c$... (5)

by (2) Art. 49

$$= x_1(ax_1 + hy_1 + g) + y_1(hx_1 + by_1 + f) + gx_1 + fy_1 + c$$

$$= gx_1 + fy_1 + c \text{ by (3)} \quad \dots \dots (6)$$

Put the values of x_1 , and y_1 from (5), Art. 49. in (6)

$$\text{then } c_1 = \frac{abc + 2fgh - af^2 - bg^2 - ch^2}{ab - h^2} \times \frac{\Delta}{ab - h^2} \quad \dots \dots (7)$$

Where Δ is called the discriminant of the general equation (1) The equation referred to the new axes through the centre is

$$ax^2 + 2hxy + by^2 + \frac{\Delta}{ab - h^2} = 0 \quad \dots \dots (8)$$

Cor. If $\Delta = 0$, then $c_1 = 0$ then Eq. (8) will represent two straight lines

Working Rule : For the reduction of the equation of a conic referred to centre as origin.

Let the equation of the conic be.

$$F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \dots \dots (1)$$

$$\text{then write } \frac{\partial F}{\partial x} = (ax + hy + g) = 0 \text{ or } ax + hy + g = 0 \quad \dots \dots (2)$$

$$\text{and } \frac{\partial F}{\partial y} = 2(hx + by + f) = 0 \text{ or } hx + by + f = 0 \quad \dots \dots (3)$$

The equations (2) and (3) i. e. $\frac{\partial F}{\partial x} = 0$, and $\frac{\partial F}{\partial y} = 0$

are the same as these giving the co-ordinates of the centre.

Now the constant $= \frac{\Delta}{ab - h^2} = gx + fy + c$ which is represented by another constant c_1 .

$$c_1 = gx + fy + c \quad \dots \dots (4)$$

where x and y are found from the equations $\frac{\partial F}{\partial x} = 0$, and $\frac{\partial F}{\partial y} = 0$

and g, f and c are found from the given equation (1)

Hence the equation reduces to $ax^2 + 2hxy + by^2 + c_1 = 0$

In other words "Same terms of second degree as in the given equation + new constant = 0.
which we can always reduce to the FORM

$$Ax^2 + 2Hxy + By^2 = 1$$

If the axes of the equation (5) are rotated about its new (centre) origin in such a way that xy -term vanishes, the new equation of the conic becomes $a_1x^2 + b_1y^2 + c_1 = 0$... (5)

which is called the **Standard Form** of the equation of the conic. Then by Invariants, (See Art. 35) we have from (5) and (6)

$$a_1 + b_1 = a + b$$

$$a_1b_1 = ab - h^2$$

where a_1 and b_1 can be easily found.

Thus the equation (1) reduces to the standard form

$$a_1x^2 + b_1y^2 + c_1 = 0$$

Ex. Reduce the equation $32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$ to the standard form.

$$\text{Let } f(x, y) \equiv 32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$$

$$\delta f / \delta x = 0 \text{ or, } 64x + 52y - 64 = 0 \quad \dots \dots \dots (1)$$

$$\text{and } \delta f / \delta y = 0 \text{ or, } 52x - 14y - 52 = 0 \quad \dots \dots \dots (2)$$

Solve (1) and (2) for x and y $\therefore x = 1, y = 0$ Centre is at $(1, 0)$

$$\text{New constant, } c_1 = gx + fy + c = -32(1) - 26(0) - 148 = -180$$

Therefore, the equation of the conic referred as origin is

$$32x^2 + 52xy - 7y^2 - 180 = 0 \text{ by (5) working Rule (xi)}$$

When the xy - term is removed by rotation of the axes, let the reduced equation be

$$a_1x^2 + b_1y^2 = 180 \dots \dots \dots (3) \text{ by (6) (cor. Art 50)}$$

$$\text{Then } a_1 + b_1 = 32 - 7 = 25.$$

$$\therefore \text{ here } a = 32, b = -7, h = 26$$

$$\text{and } a_1b_1 = -32 \cdot 7 - (26)^2 = -900$$

$$\text{Solve for } a_1 \text{ and } b_1 \therefore a_1 = 45, b_1 = -20$$

Therefore, the equation (3) is

$$45x^2 - 20y^2 = 180 \text{ or, } x^2/4 - y^2/9 = 1$$

The given equation represents a hyperbola.

Art. 51. Axes : To find the lengths and position of the axes of the conic. [C. U. 1986]

$$ax^2 + 2hxy + by^2 = 1 \quad \dots \dots \dots (1)$$

Let the axes be rectangular.

Let the conic be cut by a concentric circle of radius r . The diameters through the points of intersection will be equally inclined to the axes of the conic, and will be coincident if the radius of the circle be equal to either of the semi-axes of the conic.

$$\text{Now suppose that equation of the circle as } \frac{x^2 + y^2}{r^2} = 1 \quad \dots \dots \dots (2)$$

Making (1) homogeneous with the help of (2), we have

$$\left(a - \frac{1}{r^2}\right)x^2 + 2hxy + \left(b - \frac{1}{r^2}\right)y^2 = 0 \quad \dots \dots \dots (3)$$

The equation of the normal at (x_1, y_1) to the conic $F(x, y) = 0$ is from (2) and (5)

$$y - y_1 = \frac{hx_1 + by_1 + f}{ax_1 + hy_1 + f_1} (x - x_1)$$

$$\text{or, } \frac{y - y_1}{hx_1 + by_1 + f_1} = \frac{x - x_1}{ax_1 + by_1 + g} \quad \dots \quad (6)$$

$$\text{or, } \frac{y - y_1}{(F_y)_1} = \frac{x - x_1}{(F_x)_1} \quad \dots \quad (7)$$

where $(F_x)_1$ is the differentiation of the given equation $F(x, y) = 0$ with respect to x , treating y_1 as constant, and $(F_y)_1$ is the differentiation with respect to y_1 treating x_1 as constant.

Art. 55. Normals to different curves.

The equation to the normal at (x_1, y_1) to the curve, by (6) of Art. 54.

1. Circle $x^2 + y^2 = a^2$ is $\frac{y - y_1}{y_1} = \frac{x - x_1}{x_1}$ or, $xy_1 = x_1y$

2. Circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\frac{y - y_1}{y_1 + f} = \frac{x - x_1}{x_1 + g}$

3. Parabola, $y^2 = 4ax$ is $\frac{y - y_1}{y_1} = \frac{x - x_1}{-2a}$ or, $y - y_1 = -\frac{y_1}{2a}(x - x_1)$

4. Ellipse. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{y - y_1}{y_1/b^2} = \frac{x - x_1}{x_1/a^2}$

5. Hyperbola $x^2/a^2 - y^2/b^2 = 1$ is $\frac{y - y_1}{-y_1/b^2} = \frac{x - x_1}{x_1/a^2}$

Art. 56. Find the condition that a line may touch a conic.

Let the line be $lx + my + n = 0$ (1)

and the conic $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ (2)

If (1) touches (2) at (x_1, y_1) it must be of the form $T = 0$ the tangent at (x_1, y_1)

or, $(ax_1 + hy_1 + g)x + (hx_1 + by_1 + f)y + (gx_1 + fy_1 + c) = 0$ By 8, Art. 52.

This equation is the same as (1) if

$$\frac{ax_1 + hy_1 + g}{l} = \frac{hx_1 + by_1 + f}{m} = \frac{gx_1 + fy_1 + c}{n} = k \text{ (say)}$$

Then $ax_1 + hy_1 + g - kl = 0$ (3)

$hx_1 + by_1 + f - mk = 0$ (4)

$gx_1 + fy_1 + c - nk = 0$ (5)

Since the point (x_1, y_1) is on $lx + my + n = 0$

hence $lx_1 + my_1 + n = 0$ (6)

~~Ex~~ Reduce the equation —

$$16x^2 - 24xy + 9y^2 - 104x - 172y - 44 = 0 \quad \dots \quad (1)$$

to the standard form. Find all its properties.

As the second degree terms of (i) form a perfect square the eq.

(i) will represent a parabola since $\Delta \neq 0$ and $ab = h^2$

$$\therefore (4x - 3y)^2 = 104x + 172y - 44 \quad \dots \quad (2)$$

The lines $4x - 3y = 0$ and $104x + 172y - 44 = 0$ are not at right angles, hence let us introduce a constant λ in (2)

$$(4x - 3y + \lambda)^2 = (104x + 172y - 44) + 2\lambda(4x - 3y) + \lambda^2$$

$$\text{or, } (4x - 3y + \lambda)^2 = x(104 + 8\lambda) + y(172 - 6\lambda) + (\lambda^2 - 44)$$

$$\text{The lines } 4x - 3y + \lambda = 0 \quad \dots \quad (4)$$

$$\text{and } x(104 + 8\lambda) + y(172 - 6\lambda) + (\lambda^2 - 44) = 0 \quad \dots \quad (5)$$

are at right angles if (apply $a_1a_2 + b_1b_2 = 0$, sec Art. (25b))

$$4(104 + 8\lambda) + (-3)(172 - 6\lambda) = 0 \text{ or, } \lambda = 2 \quad \dots \quad (6)$$

Now eq. (3) becomes by (6)

$$(4x - 3y + 2)^2 = 40(3x + 4y - 1)$$

which can be written as

$$\left(\frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} \right)^2 \cdot (25) = 40 \cdot 5 \left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} \right)$$

$$\text{or, } PN^2 = \frac{40 \cdot 5}{25} PM \text{ or, } Y^2 = 8X$$

Which is the standard form of eq. (1)

$$\text{or, } Y^2 = 4PX. \text{ Where } P = \frac{1}{4} \cdot 8 = 2$$

$$\text{For other properties of } Y^2 = 8x = 4pX \quad \dots \quad (7)$$

$$\text{Here } Y = \frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} \text{ and } X = \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} \quad \dots \quad (8)$$

$$(i) \text{ Axis of (7) is } Y = 0 \text{ or, } 4x - 3y + 2 = 0, \text{ by (8)} \quad \dots \quad (9)$$

$$(ii) \text{ Tangent at vertex is } X = 0, \text{ or, } 3x + 4y - 1 = 0 \text{ by (8)} \quad \dots \quad (10)$$

(iii) Vertex is the intersection of eq. (9) and (10) by (8)

i. e. $4x - 3y + 2 = 0$ and $3x + 4y - 1 = 0$, Solve.

$$x = -\frac{1}{5}, y = \frac{2}{5} \text{ Vertex } \left(-\frac{1}{5}, \frac{2}{5} \right)$$

(iv) Latus rectum is $4p = 8$

Now compare the co-efficients of (3) and (4)

$$7A + B = -170, -4A + 8B = 140, A = -25, B = 5$$

Asymptotes from (4) are

$$x + 8y - 25 = 0 \text{ and } 7x - 4y + 5 = 0$$

Ex 9. Reduce the equation

$$4x^2 - 4xy + y^2 - 8x - y + 6 = 0$$

to the standard form. Find all its properties.

In the eq. (1), $a = 4, b = 1, h = -2, g = -4, f = -3, c = 5$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = -64, ab - h^2 = 4 - 4 = 0$$

$\Delta \neq 0$ and $ab = h^2$. Hence eq. (1) represents a parabola. The eq. (1) may be written as

$$(2x - y)^2 = 8x + 6y - 5$$

$$\text{or, } (2x - y + \lambda)^2 = 8x + 6y - 5 + \lambda^2 + 2\lambda(2x - y)$$

$$\text{or, } (2x - y + \lambda)^2 = x(4\lambda + 8) + y(-2\lambda + 6) + (\lambda^2 - 5)$$

$$\text{The lines } 2x - y + \lambda = 0$$

$$\text{and } x(4\lambda + 8) + y(-2\lambda + 6) + (\lambda^2 - 5) = 0$$

may be perpendicular if $a_1a_2 + b_1b_2 = 0$

$$\text{i. e. } 2(4\lambda + 8)(-1) + (-2\lambda + 6) = 0 \text{ or, } \lambda = -1$$

The eq. (3) becomes

$$(2x - y - 1)^2 = 4(x + 2y - 1) \text{ which can be written as}$$

$$\left(\frac{2x - y - 1}{\sqrt{(2^2 + 1^2)}} \right)^2 = 4\sqrt{5} \left(\frac{x + 2y - 1}{\sqrt{(1^2 + 2^2)}} \right)$$

It is of the form $PN^2 = 4(\sqrt{5})/5$. PM.

or, $Y^2 = 4/\sqrt{5}$. X. which is the standard equation of a parabola.

$$(i) \text{ Latus rectum} = 4p = 4/\sqrt{5} \text{ or, } p = 1/\sqrt{5}$$

$$X = (x + 2y - 1)/\sqrt{5} \text{ and } Y = (2x - y - 1)/\sqrt{5}$$

$$\therefore Y^2 = 4pX.$$

$$(ii) \text{ Axis is } Y = 0 \text{ or, } 2x - y - 1 = 0 \text{ from (7) ...}$$

$$(iii) \text{ Tangent at the vertex } X = 0 \text{ i. e. } x + 2y - 1 = 0 \text{ from (7) ...}$$

$$(iv) \text{ Vertex A is the intersection of eqs. (8) and (9).}$$

$$\text{Solve } x = 3/5, y = 1/5; A(3/5, 1/5)$$

$$(v) \text{ Focus is } S(p, 0) \text{ i.e. ; } X = p \text{ and } Y = 0$$

$$\text{or, } \frac{x + 2y - 1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \text{ and } \frac{2x - y - 1}{\sqrt{5}} = 0 \text{ from (7)}$$

$$\text{or, } x + 2y - 2 = 0 \text{ and } 2x - y - 1 = 0$$

$$\text{Solve, } x = 4/5, y = 3/5; S(4/5, 3/5)$$

[D.U.(P) 1986 (C. H. 1977)]

(1)

(2)

(3)

(4)

(5)

(6)

(7)

(8)

(9)

(vi) Equation of latus rectum is $X = p$

or, $(x + 2y - 1)/\sqrt{5} = 1/\sqrt{5}$ or, $x + 2y - 2 = 0$

(vii) Foot of the directrix $Z(-p, 0)$ i. e. $X = -p, Y = 0$

or, $(x + 2y - 1)/\sqrt{5} = -1/\sqrt{5}$ and $(2x - y - 1)/\sqrt{5} = 0$

Solve, $x = 4/5, y = -1/5 \therefore Z(4/5, -1/5)$.

(viii) Equation of the directrix is $X = -p$

or, $(x + 2y - 1)/\sqrt{5} = -1/\sqrt{5}$ or, $x + 2y = 0$

Ex. 10. Reduce the equation $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ to the standard form. Find also the equations of latus rectum, directrices and axes. [N. U. H. 1997]

Let $f(x, y) \equiv x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$... (1)

(i) its centre is at $\left(0, \frac{1}{2}\right)$ [See Ex. 4. for method.]

Now transfer the origin to the point $\left(0, \frac{1}{2}\right)$, the eq. (1) reduces to $x^2 + 12xy - 4y^2 + c_1 = 0$

where $c_1 = gx_1 + fy_1 + c = -3 \cdot 0 + 2 \cdot \frac{1}{2} + 9 = 10$

(ii) The reduced equation is $x^2 + 12xy - 4y^2 + 10 = 0$

or, $-x^2/10 - 12xy/10 + 4y^2/10 = 1$

(iii) The lengths of the semi axes are given by

$$(a - 1/r^2)(b - 1/r^2) = h^2 \text{ or, } (-1/10 - 1/r^2)(4/10 - 1/r^2) = (-6/10)^2$$

$$\text{or, } r^2 = 5/4 \text{ or, } -2, \therefore r_1^2 = 5/4 : r_2^2 = -2 \quad \dots \quad (2)$$

The conic is a hyperbola.

(iv) Eq. of the transverse axis is

$$(a - 1/r_1^2)x + hy = 0 \text{ or, } (-1/10 - 4/5)x - 3/5y = 0 \text{ or, } 3x + 2y = 0$$

referred to the centre $\left(0, \frac{1}{2}\right)$ or, $3(x - 0) + 2(y - 1/2) = 0$

referred to the origin, or, $3y + 2y - 1 = 0 \quad \dots \quad (3)$

Slope of it, $\tan \theta = -3/2$. $\sin \theta = 3/\sqrt{13}$, $\cos \theta = -2/\sqrt{13} \quad \dots \quad (4)$

through $\left(0, \frac{1}{2}\right)$, then $k = 3/2$, The equation is $4x - 6y + 3 = 0$

(v) Eccentricity. $e^2 = 1 - r_2^2/r_1^2 = 1 + 2.4/5$... (5)

or, $e = \sqrt{13/5}$

(vi) d for foci S and $S' = \pm ae = \pm r_1 e = \pm \frac{1}{2}\sqrt{13}$

d for feet of the directrices Z and $Z' = \pm a/e = \pm 5/2\sqrt{13}$

(iv) Latus rectum is $4p = 8$

(v) Focus $X = p, Y = 0$, by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = 2 \text{ and } \frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} = 0$$

$$\text{or, } 3x + 4y - 11 = 0 \text{ and } 4x - 3y + 2 = 0. \text{ solve}$$

$\therefore x = 1, y = 2$. The focus is $(1, 2)$

(vi) Foot of the directrix, $X = -p$ and $Y = 0$ by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = 2 \text{ and } \frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} = 0$$

$$\text{or, } 3x + 4y + 9 = 0 \text{ and } 4x - 3y + 2 = 0. \text{ Now solve. The foot of the directrix is } (-7/5, -6/5)$$

(vii) Equation of the directrix, $X = -p$, by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = -2 \text{ or, } 3x + 4y + 9 = 0$$

(viii) Equation of the latus rectum $X = p$ by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = 2 \text{ or, } 3x + 4y - 11 = 0$$

Note : "The properties of a parabola can be determined by other method also. The demonstration of such a method is given below.

Ex. 14(a) What conic does the equation $5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0$ represent? Reduce the equation to standard form. [N.U.H. 1996]

Ans. The given equation

$$5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0 \quad \dots \quad \dots \quad \dots \quad (i)$$

Comparing (i) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

We get, $a = 5, h = -1, b = 5, g = -4, f = -4, c = -8$.

$$\therefore \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 5.5. -8 + 2(-4)(-4)(-1) - 5(-4)^2 - 5(-4)^2 - (-8)(-1)^2$$

$$= -200 - 32 - 80 - 80 + 8 = -384$$

$$\therefore \Delta \neq 0$$

$$ab - h^2 = 5.5 - (-1)^2 = 25 - 1 = 24$$

$$\therefore ab - h^2 > 0$$

Thus, the given conic represents an ellipse,

Let (α, β) be the centre of the conic (i)

$$\therefore \alpha = \frac{hf - bg}{ab - h^2},$$

$$\beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{(-1)(-4) - 5(-4)}{5.5 - (-1)^2}$$

$$= \frac{24}{24}$$

$$= 1$$

$$= \frac{(-1)(-4) - 5(-4)}{5.5 - (-1)^2}$$

$$= \frac{24}{24}$$

$$= 1$$

∴ Centre is at (1, 1)

Transfer the origin to a point (1,1) When direction of axes are unchanged, then (1) becomes

$$ax^2 + 2hxy + by^2 = c_1$$

$$\text{ie } 5x^2 - 2xy + 5y^2 = c_1$$

$$\text{where } c_1 = -(\alpha g + \beta f + c)$$

$$c_1 = -\{1 \cdot (-4) + 1 \cdot (-4) - 8\} = 16$$

$$\therefore 5x^2 - 2hxy + 5y^2 = 16$$

$$\Rightarrow \frac{5}{16}x^2 - 2 \cdot \frac{1}{16}xy + \frac{5}{16}y^2 = 1$$

$$\Rightarrow Ax^2 + 2Hxy + By^2 = 1 \dots \dots \dots (2)$$

$$\text{Where } A = \frac{5}{16}, H = \frac{-1}{16}, B = \frac{5}{16}$$

Removing xy term when the axes being rotated.

$$\therefore a_1x^2 + b_1y^2 = 16 \dots \dots \dots (3)$$

$$a_1 + b_1 = a + b$$

$$a_1b_1 = ab - h^2$$

$$\therefore a_1 + b_1 = 5 + 5 = 10$$

$$a_1b_1 = 5.5 - (-1)^2$$

$$a_1b_1 = 24$$

$$\begin{aligned} \text{Now, } (a_1 - b_1)^2 &= (a_1 + b_1)^2 - 4a_1b_1 \\ &= 10^2 - 4 \cdot 24 = 100 - 96 = 4 \end{aligned}$$

$$\therefore a_1 - b_1 = 2 \text{ take +ve sign}$$

$$\therefore \begin{aligned} a_1 + b_1 &= 10 \\ a_1 - b_1 &= 2 \end{aligned}$$

$$\hline \begin{aligned} (+) \quad 2a_1 &= 12 \end{aligned}$$

$$b_1 = 10 - a_1 = 10 - 6 = 4$$

$$\therefore (3) \Rightarrow 6x^2 + 4y^2 = 16$$

$$\Rightarrow \frac{x^2}{\frac{8}{3}} + \frac{y^2}{4} = 1$$

$$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{8}{3}}\right)^2} + \frac{y^2}{4} = 1$$

$$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{8}{3}}\right)^2} + \frac{y^2}{4} = 1$$

which is standard form of an ellipse.

Ex. 14(b) Find the asymptotes the hyperbola $x^2 + 2xy - y^2 + 2x + 4y = 0$ and the equation of the conjugate hyperbola. [N.U.H. 1996]

Ans. The given hyperbola

$$x^2 + 2xy - y^2 + 2x + 4y = 0 \dots (1)$$

Let equation of asymptote of the hyperbola (1) is

$$x^2 + 2xy - y^2 + 2x + 4y + \lambda = 0 \dots (2)$$

Since asymptotes is a pair of st. lines

$$\text{So } \Delta = 0 \text{ is } \boxed{abc + 2fgh - af^2 - bg^2 - ch^2 = 0}$$

$$\therefore 1. (-1). \lambda + 2. (+2)(+1). -1. (+2)^2 - (-1)(+1)^2 - \lambda. 1^2 = 0$$

$$\Rightarrow -\lambda + 4 - 4 + 1 - \lambda = 0$$

$$\Rightarrow 2\lambda + 1 = 0$$

$$\therefore \lambda = -\frac{1}{2}$$

$$\therefore (2) \Rightarrow x^2 + 2xy - y^2 + 2x + 4y + \frac{1}{2} = 0$$

Which is the required asymptotes.

Let equation of the conjugate hyperbola is

$$x^2 + 2xy - y^2 + 2x + 4y + k = 0 \dots (3)$$

We know, equation of the conjugate hyperbola

$$= 2 \times \{\text{equation of asymptotes}\} - \text{equation of the hyperbola}$$

$$\Rightarrow x^2 + 2xy - y^2 + 2x + 4y + k = 2 \times (x^2 + 2xy - y^2 + 2x + 4y + \frac{1}{2}) - (x^2 + 2xy - y^2 + 2x + 4y)$$

$$\Rightarrow k = 1$$

$$(3) \Rightarrow x^2 + 2xy - y^2 + 2x + 4y + 1 = 0$$

Which is the required conjugate hyperbola.

Ex. 14(c) Reduce the equation $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ to the standard form and identify it. [N.U.H. 1997]

Ans. The given equation

$$x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0 \dots (1)$$

Compare (1) with the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ We get}$$

$$a = 1, b = -4, h = 6, g = -3, f = 2, c = 9$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 1. (-4). 9 + 2. 2. (-3). 6 - 1. 2^2 + 4. (-3)^2 - 9. 6^2$$

$$= -36 - 72 - 4 + 36 - 324$$

$$= -400 \neq 0$$

$$ab - h^2 = 1. -4 - 6^2 = -40 < 0$$

Thus, the given conic represents a hyperbola.

Let (α, β) be the centre of the conic (1)

$$\therefore \alpha = \frac{hf - bg}{ab - h^2} \quad \beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{6 \cdot 2 + 4(-3)}{1 - 4 - 6^2} = \frac{6 - 3 - 1.2}{1 - 4 - 6^2}$$

$$= \frac{0}{-40} = \frac{-20}{-40}$$

$$= 0 = \frac{1}{2}$$

$\therefore$ Centre is at $(0, \frac{1}{2})$

Now, transfer the origine to a point $(0, \frac{1}{2})$

Then (1) becomes

$$x^2 + 12xy - 4y^2 = c_1 \dots\dots (2)$$

Where $C_1 = -(g\alpha + f\beta + C)$

$$= -(-3 \cdot 0 + 2 \cdot \frac{1}{2} + 9)$$

$$= -10$$

$$\Rightarrow x^2 + 12xy - 4y^2 = -10 \dots\dots (3)$$

Now, transform the axes such that remove the xy term, then

$$a_1 x^2 + b_1 y^2 = -10 \dots\dots (4)$$

Where $a_1 b_1 = a + b = 1 - 4 = -3$

and $a_1 b_1 = ab - h^2 = -4 - 6^2 = -40$

Now, $(a_1 - b_1)^2 = (a_1 + b_1)^2 - 4 a_1 b_1$

$$= (-3)^2 - 4 \cdot -40$$

$$= 9 + 160 = 169$$

$\therefore a_1 - b_1 = 13$ Take + ve sign

Again, $a_1 + b_1 = 3$

$$\begin{array}{rcl} a_1 - b_1 & = & 13 \\ (+) 2a_1 & = & 10 \end{array} \therefore a_1 = 5$$

$$b_1 = -3 - a_1 = -3 - 5 = -8$$

$$(4) \Rightarrow 5x^2 - 8y^2 = -10$$

$$\Rightarrow \frac{x^2}{-2} + \frac{y^2}{\frac{5}{4}} = 1$$

Which is required standard form.

Ex. 14(d) Find the nature of the Conic $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$. Reduce it to standard form. Find vertex, Focus, equation and length of latus rectum of the Conic. [N.U.H - 1999]

Ans. The given equation

$$x^2 + 2xy + y^2 - 6x - 2y + 4 = 0 \dots\dots\dots (1)$$

Compare (1) with the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \text{we get}$$

$$a = 1, h = 1, b = 1, g = -3, f = -1, c = 4$$

$$\begin{aligned} \therefore \Delta &= abc + 2fgh - af^2 - by^2 - ch^2 \\ &= 1.1.4 + 2.(-1)(-3).1 - 1.(-1)^2 - 1(-3)^2 - 4.1^2 \\ &= 4 + 6 - 1 - 9 - 4 = -4 \end{aligned}$$

$$\therefore \Delta \neq 0$$

$$ab - h^2 = 1.1 - 1^2 = 0$$

Thus, the given conic represents a parabola

(1) can be written as

$$(x + y)^2 = 6x + 2y - 4$$

$$\Rightarrow (x + y + \lambda)^2 = 6x + 2y - 4 + 2\lambda x + 2\lambda y + \lambda^2$$

$$\Rightarrow (x + y + \lambda)^2 = (6 + 2\lambda)x + (2 + 2\lambda)y + \lambda^2 - 4 \dots\dots\dots (2)$$

Find the value of λ , let the lines

$$x + y + \lambda = 0 \text{ and } (6 + 2\lambda)x + (2 + 2\lambda)y + \lambda^2 - 4 = 0$$

are perpendicular $[a_1 a_2 + b_1 b_2 = 0]$

$$\therefore 1. (6 + 2\lambda) + 1. (2 + 2\lambda) = 0$$

$$\Rightarrow 6 + 2\lambda + 2 + 2\lambda = 0$$

$$\Rightarrow 4\lambda = -8 \therefore \lambda = -2$$

$$(2) \Rightarrow (x + y - 2)^2 = (6 - 4)x + (2 - 4)y + (-2)^2 - 4$$

$$\Rightarrow (x + y - 2)^2 = 2x - 2y$$

$$\Rightarrow \left(\frac{x+y-2}{\sqrt{1^2+1^2}} \right)^2 = \frac{2(x-y)}{\sqrt{1^2+1^2}} \cdot \sqrt{2}$$

$$\Rightarrow \left(\frac{x+y-2}{\sqrt{2}} \right)^2 = 4 \cdot \frac{\sqrt{2}}{4} \cdot \left(\frac{x-y}{\sqrt{2}} \right)$$

$$\therefore Y^2 = 4AX. \text{ Where } Y = \frac{x+y-2}{\sqrt{2}}, X = \frac{x-y}{\sqrt{2}}$$

$$A = \frac{\sqrt{2}}{4}$$

Which is required standard form.

For Vertex $X = 0, Y = 0$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = 0 \quad \Rightarrow \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow x - y = 0 \quad \Rightarrow x + y - 2 = 0$$

SOLID GEOMETRY

[THREE DIMENSIONAL GEOMETRY]

RECTANGULAR CO-ORDINATES

1

Art. 1. Introduction : The position of a point in space may be defined by referring it to three fixed planes. These planes have a common point, known as the **Origin** and intersect in pairs in three straight lines, known as **Co-ordinate Axes**. The planes themselves are called the **Co-ordinate planes**. When the three Co-ordinate planes and consequently the three co-ordinate axes are mutually at right angles, the axes are said to be **Rectangular**. The three planes divide the space into eight parts known as the **Octants**.

Art. 2. Co-ordinates of a point in space : Let P be any point in space. Through P draw a plane $PMAN$ parallel to yz plane and hence perpendicular to XOX' cutting it in the point A , then $OX = x$ is called the x co-ordinate of the point P . Again through P draw another plane $PNBL$ parallel to zx plane and hence perpendicular to YOY' cutting it in B then $ON = y$ is called the y -co-ordinate of the point P . Similarly $OC = z$ is the z -co-ordinate of the point P where C is the

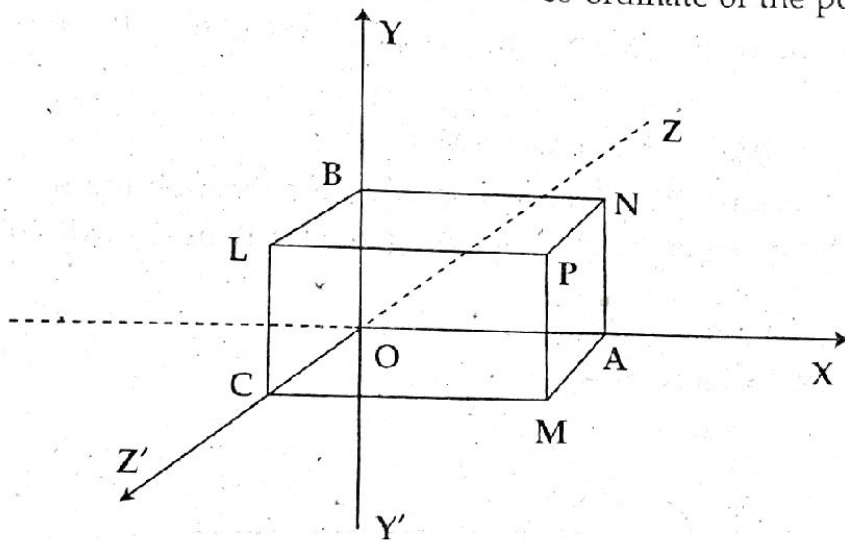


Fig: 1

point in which the plane $PLCM$ through P parallel to xy plane cuts the z -axis ZOZ' .

The three numbers x, y, z are called the **co-ordinates of the point P** and the point is represented as $P(x, y, z)$.

To determine the position of any point P it is not sufficient merely to know the absolute lengths of the lines LP, MP, NP we must also know the directions in which they are drawn. If lines drawn in one directions be considered as positive, those drawn in the opposite direction must be considered negative. We shall consider the direction as OX, OY, OZ are positive.

The three co-ordinate planes divide the whole of the space into eight compartments which are called **octants**. The octant $OXYZ$ in which the point P lies is called **first octant** and any point in this has each of its co-ordinates positive. Again since each of the co-ordinates of

Thus $ABCD$ is a tetrahedron having for triangular faces, the Δ 's ABC , ACD , ADB , BCD . The points A, B, C, D , are called the VERTICES of the tetrahedron. The edges AB and CD ; BC and AD ; CA and BD which do not meet are called the pairs of opposite edges of the tetrahedron.

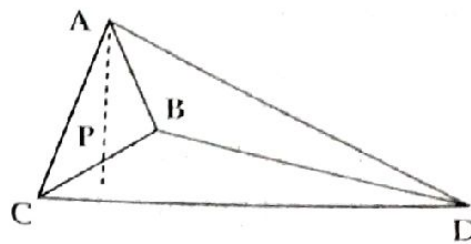


Fig: 5

Art. 7. To prove that the centre of gravity of a tetrahedron the co-ordinates of whose vertices are (x_r, y_r, z_r) where $r = 1, 2, 3, 4$ is the point.

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

Let A, B, C, D be the points (x_1, y_1, z_1) ; (x_2, y_2, z_2) ; (x_3, y_3, z_3) ; (x_4, y_4, z_4) . If G_1 be the centre of gravity of the triangular face BCD , then the point G_1 is

$$\left(\frac{x_2 + x_3 + x_4}{3}, \frac{y_2 + y_3 + y_4}{3}, \frac{z_2 + z_3 + z_4}{3} \right)$$

Now we know from Statics that the C. G of a tetrahedron is on the line joining any vertex to the C. G of the opposite face and divides it in the ratio 3 " 1.

Hence if $G(x, y, z)$ be the C. G of the tetrahedron; then

$$x = \frac{3 \cdot \frac{x_2 + x_3 + x_4}{3} + 1 \cdot x_1}{4} = \frac{x_1 + x_2 + x_3 + x_4}{4}$$

$$\text{Similarly } y = \frac{x_1 + x_2 + x_3 + x_4}{4}, z = \frac{y_1 + y_2 + y_3 + y_4}{4} \dots \quad (8)$$

The C. G of the tetrahedron is therefore,

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

Art. 8. Direction Cosines of a Line.

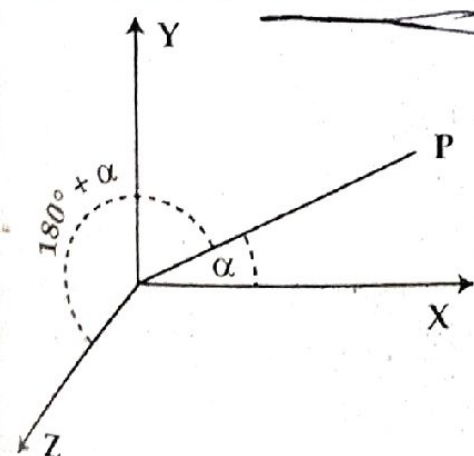


Fig: 6

(If a given line OP makes angles α, β, γ , with the positive direction of axes of x, y and z respectively then $\cos \alpha, \cos \beta, \cos \gamma$ are the Direction Cosines of the line OP and are generally denoted by the letters, l, m, n respectively) Again the direction cosines of PO will be $\cos(180 + \alpha), \cos(180 + \beta), \cos(180 + \gamma)$ because the line through O , i. e., OP will now make angles $180 + \alpha, 180 + \beta,$

$180 + \gamma$ with the axes.

The angles α, β, γ , of the line OP are called the Direction Angle of OP .

Note : If a line does not pass through the origin O , then its direction angles are the angles between the axes and a line drawn through O parallel to the given line (and having the same direction.)

Cor. Direction cosines (d, c, s) of the Axes of References : Now the axis of x makes with the co-ordinates axes the angles $0^\circ, 90^\circ, 90^\circ$, respectively and hence by definition its direction cosines are $\cos 0^\circ, \cos 90^\circ, \cos 90^\circ$, i. e. $(1, 0, 0)$. Similarly the d, c, s of the axes of y and z are $(0, 1, 0)$ and $(0, 0, 1)$ respectively.

Art. 9. A Useful Relation : If O be the origin and (x, y, z) the co-ordinates of a point P , then $x = lr, y = mr, z = nr$

Where l, m, n , are direction cosines of OP and r , is the length of OP .

Through p draw PL perpendicular to x -axis so that $OL = x$. From the right angled triangle OPL , we have

$$\frac{OL}{OP} = \cos \angle LOP = \cos \alpha \text{ or, } \frac{x}{r} = l \text{ or, } x = lr$$

Similarly we have $y = mr, z = nr$,

Therefore the co-ordinates of P are $x = lr, y = mr, z = nr$...

Note : $\frac{x}{l} = \frac{y}{m} = \frac{z}{n} = r$...

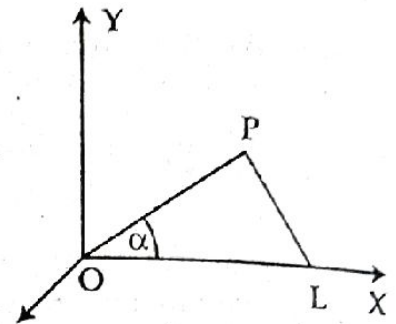


Fig: 7

(9)

9(a)

Art. 10. Relation between direction cosines. To prove that if l, m, n , be the direction cosines of any line then $l^2 + m^2 + n^2 = 1$, i. e. the sum of the squares of the direction cosines of every line is one.

Let OP be drawn through the origin parallel to the given line so that l, m, n , are the cosines of angles which makes with OX, OY, OZ respectively.

Let (x, y, z) be the co-ordinates of any point P on this line

Let $OP = r$

$$\therefore x = lr, y = m. r, z = nr$$

(By eq, 9 of Art. 9)

Squaring and adding, we obtain

$$x^2 + y^2 + z^2 = (l^2 + m^2 + n^2) r^2$$

$$\text{or, } r^2 = (l^2 + m^2 + n^2) r^2$$

(By eq (2) of Art. (5))

$$\therefore l^2 + m^2 + n^2 = 1$$

$$\text{or, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

(10)

[R. U. 1980]

Art. 11. Direction Ratios : Any three numbers a, b , and c which are proportional to the direction cosines l, m, n , respectively of a given line are called the direction ratios of the given line (Rectangular axes). Now from definition of direction ratios.

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}} = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}} \quad (\text{By eq. 10 of art. 10})$$

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

Where the same sign either + ve or -ve is to be chosen throughout.

Hence of (a, b, c) , be three numbers proportional to the d.c's of OP , then the actual direction cosines of OP are

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}} \quad \dots \quad (11)$$

and that of PO are

$$\frac{-a}{\sqrt{a^2 + b^2 + c^2}}, \frac{-b}{\sqrt{a^2 + b^2 + c^2}}, \frac{-c}{\sqrt{a^2 + b^2 + c^2}} \quad \dots \quad (12)$$

Art. 12. Projection of a point on a straight line.

The projection of a point A on a given line CD is the point B which is the foot of the perpendicular from A on CD .

This the point in which the plane through A and perpendicular to CD meets it.

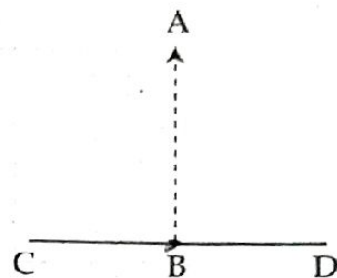


Fig : 8

Art. 13. The orthogonal projection of a given segment AB of a line on any line CD is $A'B'$ where θ is the angle between AB and CD .

Let the planes through A and B perpendicular to the line CD meet it in A', B' respectively so that A', B' is the projection of AB on CD .

Through A draw a line AP parallel to CD to meet the plane through B at P .

Now AP is parallel to CD ; $\angle PAB = \theta$

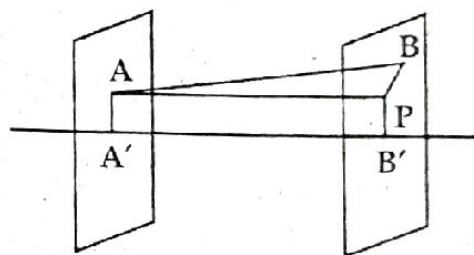


Fig : 9

Also PB lies in the plane which is perpendicular to AP .

$$\angle APB = 90^\circ$$

$$\text{Hence } AP = AB \cos \theta$$

Clearly $A'B'PA$ is a rectangle so that we have $AP = A'B'$

$$\text{Hence } A'B' = AB \cos \theta$$

Art. 14. Direction cosines of a line joining two point $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

Let L, M be the feet of the perpendiculars drawn from P, Q to the X -axis respectively so that $OL = x_1, OM = x_2$

$$\text{Projection of } PQ \text{ on } X \text{ axis} = LM = OM - OL = x_2 - x_1$$

Also if l, m, n be the direction cosines of PQ the projection of PQ on X -axis $= l PQ$.

Similarly pojection of PQ on Y - axis abd Z - axis, are $m PQ = y_2 - y_1, n PQ = z_2 - z_1$

$$\therefore \frac{x_2 - x_1}{l} = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n} = PQ \quad \dots \quad (13)$$

This the direction cosines of the line joining the two points (x_1, y_1, z_1) and (x_2, y_2, z_2) are

$$x_2 - x_1; y_2 - y_1; z_2 - z_1 \quad \dots \quad (14)$$

Art. 15. Projection of a broken line consisting of continuous segments on a given line.

Let $P_1, P_2, P_3, \dots, P_n$ be any number of points in space and $Q_1, Q_2, Q_3, \dots, Q_n$ be their projections on the line AB , and these points being on the same straight line, we have

$Q_1Q_n = \text{Projection of } P_1P_n = Q_1Q_2 + Q_2Q_3 + Q_3Q_4 + \dots + Q_{n-1}Q_n$. Where $Q_1Q_2 = \text{Projection of } P_1P_2$

$Q_2Q_3 = \text{Projection of } P_2P_3$ and so on

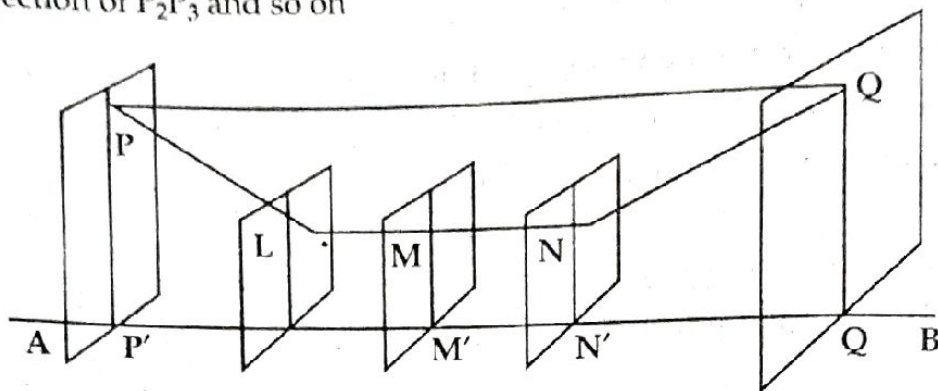


Fig : 10

Hence the sum of the projection of $P_1P_2, P_2P_3, \dots, P_{n-1}P_n$ on $AB = \text{Projection of } P_1P_n$ on $AB = Q_1Q_n$... (15)

Art. 16. Projection of the join of two points on a line :

To show that the projection of the line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on a line with direction cosines l, m, n is $(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n$.

Through PQ draw lines parallel to the co-ordinates planes to form rectangular parallelopiped whose one diagonal is PQ (Fig. 3)

Now $PA = x_2 - x_1, AN = y_2 - y_1, NQ = z_2 - z_1$

The lines PA, AN, NQ are respectively parallel to x -axis, y -axis and z -axis. Therefore their respective projections on the line with direction cosines l, m, n are

$$(x_2 - x_1)l, (y_2 - y_1)m, (z_2 - z_1)n$$

As the projection of PQ on any two lines is equal to the sum of the projection of PA, AN, NQ on that line therefore the required projection is $(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n$... (16)

Art. 17. Angle between two lines : If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines AB and CD and θ be the angle between them, then

$$\cos \theta = l_1l_2 + m_1m_2 + n_1n_2$$

First Method : Let OP_1, OP_2 be lines through the origin parallel to the given lines so that the cosines of the angles which OP_1 and OP_2 make with the axes are l_1, m_1, n_1 and l_2, m_2, n_2 respectively and the angle between the line is the angle between OP_1 and OP_2 . Let the angle be θ .

The projection of the line OP_2 joining $O(0, 0, 0)$ and $P_2(x_2, y_2, z_2)$ on the line OP_1 whose direction cosines are, l_1, m_1, n_1 is $(x_2 - 0)l_1 + (y_2 - 0)m_1 + (z_2 - 0)n_1$

$$s = l_1x_2 + m_1y_2 + n_1z_2$$

...

...

...

(1)

by Lagranges Identity

$$\text{or, } \sin^2 \theta = \left| \begin{array}{ccc} l_1 m_1^2 & & \\ & m_1 n_1^2 & \\ & & n_1 l_1^2 \end{array} \right| + \left| \begin{array}{ccc} l_2 m_2^2 & & \\ & m_2 n_2^2 & \\ & & n_2 l_2^2 \end{array} \right| + \dots \dots \dots (20)$$

$$\text{or, } \sin \theta = \pm \sqrt{(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2}$$

$$\begin{aligned} \text{Again } \tan \theta &= \frac{\sin \theta}{\cos \theta} \\ &= \pm \frac{\sqrt{(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2}}{(l_1 l_2 + m_1 m_2 + n_1 n_2)} \dots \dots \dots (21) \end{aligned}$$

Art. 17. (b) Corresponding formula when direction ratios of the lines are given.

If a_1, b_1, c_1 and a_2, b_2, c_2 be proportional to the direction cosines of the lines OP_1 and OP_2 then their actual direction cosines are

$$\left\{ \frac{\pm a_1}{\sqrt{\sum a_1^2}}, \frac{\pm b_1}{\sqrt{\sum a_1^2}}, \frac{\pm c_1}{\sqrt{\sum a_1^2}} \right\} \text{ and } \left\{ \frac{\pm a_2}{\sqrt{\sum a_2^2}}, \frac{\pm b_2}{\sqrt{\sum a_2^2}}, \frac{\pm c_2}{\sqrt{\sum a_2^2}} \right\}$$

If θ be the angle between them, then

$$\cos \theta = \frac{\pm (a_1 a_2 + b_1 b_2 + c_1 c_2)}{\sqrt{(\sum a_1^2)} \sqrt{(\sum a_2^2)}} \dots \dots \dots (22)$$

$$\text{and } \sin \theta = \pm \frac{\sqrt{(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}} \dots \dots \dots (23)$$

Art. 17. (c) Condition for perpendicularity of two lines.

If the two lines OP_1 and OP_2 are perpendicular to each other, then $\cos \theta = 0$ as $\theta = 90^\circ$ then the relation becomes.

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0 \dots \dots \dots (24)$$

If we use direction ratios, then we get the required condition from eq (22)

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0 \dots \dots \dots (25)$$

Art. 17. (d) Condition for parallelism of two lines.

If the two lines OP_1 and OP_2 are parallel to each other, then $\sin \theta = 0$, as $\theta = 0^\circ$

From eq (19) we have the condition

$$(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 = 0$$

Hence we have, $l_1 m_2 - l_2 m_1 = 0$; $m_1 n_2 - m_2 n_1 = 0$ and $n_1 l_2 - n_2 l_1 = 0$

$$\text{or, } \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{\sqrt{(l_1^2 + m_1^2 + n_1^2)}}{\sqrt{(l_2^2 + m_2^2 + n_2^2)}} = 1$$

$$l_1 = l_2; m_1 = m_2; n_1 = n_2 \dots \dots \dots (26)$$

i. e. The Direction Cosines of two lines are equal. From the eq (23).

we have

$$(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2 = 0$$

$$b_1 c_2 - b_2 c_1 = 0; c_1 a_2 - c_2 a_1 = 0; a_1 b_2 - a_2 b_1 = 0$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \dots \dots \dots (27)$$

The direction ratios of the two lines are proportional.

EXAMPLES

Ex. 1. Find the distance between the points $P_1 (4, 3, -6)$ and $P_2 (-2, 1, -3)$
 $\therefore P_1 P_2^2 = (4 + 2)^2 + (3 - 1)^2 + (-6 + 3)^2 = 49$
 $\therefore P_1 P_2 = 7$ Ans.

Ex. 2. A, B, C are three points on the axes x , y , and z respectively at distance a , b , c from origin O , find the co-ordinates of the point which is equidistant from A, B, C and O.

Let P be the required point (x, y, z) and the points A, B, C and O are $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ and $(0, 0, 0)$ we are given that $PO = PA = PB = PC$

Taking $PO = PA$ or, $PO^2 = PA^2$, we have

$$x^2 + y^2 + z^2 = (x - a)^2 + y^2 + z^2$$

$$\text{or, } 2ax = a^2 \text{ or } x = a/2$$

Similarly taking $PO^2 = PB^2$ and $PO^2 = PC^2$ we have

$$y = b/2 \text{ and } z = c/2$$

Hence the required point P is $\left(\frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}c\right)$

Ex. 3. Find the ratio in which the yz plane divides the joint of the points $(-2, 4, 7)$ and $(3, -5, 8)$ and also find the co-ordinates of the point of intersection of this line with the yz plane.

The co-ordinates of any point on the line joining the two points are

$$\left(\frac{3\lambda - 2}{\lambda + 1}, \frac{-5\lambda + 4}{\lambda + 1}, \frac{8\lambda + 7}{\lambda + 1}\right)$$

If the points in the yz -plane, then its x co-ordinate should be zero.

$$\frac{3\lambda - 2}{\lambda + 1} = 0 \text{ or, } 3\lambda - 2 = 0 \text{ or, } \lambda = \frac{2}{3}$$

Hence the required ratio is $2 : 3$ and putting $\lambda = 2/3$ the required point is $(0, 2/5, 37/15)$

Ex 4. Find the direction cosines of the line drawn from the origin to the point $(-6, 2, 3)$

$$\begin{aligned} \cos \theta &= \frac{x_2 - x_1}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}} \\ &= \frac{-6 - 0}{\sqrt{(-6 - 0)^2 + (2 - 0)^2 + (3 - 0)^2}} = \frac{6}{7} \end{aligned}$$

Similarly, $\cos \beta = 2/7$ and $\cos \gamma = 3/7$

The direction cosines are $-6/7, 2/7, 3/7$

Ex. 5. Find the direction cosines of the line which is equally inclined to the axes.

If the line makes angles α, β, γ with the axes then

$$\cos \alpha = \cos \beta = \cos \gamma \therefore \alpha = \beta = \gamma \text{ (given)}$$

$$\text{or } \frac{l}{1} = \frac{m}{1} = \frac{n}{1} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{1 + 1 + 1}} = \pm \frac{1}{\sqrt{3}}$$

$$\therefore d, c, s \text{ are } \left(\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}\right)$$

Ex 13 Find the distance

equal angles with the axes.

DA(1, -2, 3)

from the line

PQ through P(2, -3, 5) and Q(2, -3, 5)

which makes

The direction cosines of PQ are $1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}$

If p be the required perpendicular distance then

$$p^2 = \left\{ \frac{(-3+2)}{\sqrt{3}} - \frac{(5-3)}{\sqrt{3}} \right\}^2 + \left\{ \frac{(5-3)}{\sqrt{3}} - \frac{(2-1)}{\sqrt{3}} \right\}^2 + \left\{ \frac{(2-1)}{\sqrt{3}} - \frac{(3+2)}{\sqrt{3}} \right\}^2$$

$$= 1/3 \cdot 14 \text{ or } p = \sqrt{14/3} \text{ (see Art 33)}$$

direction cosines or given by the ... Find ... angle between the lines whose

Solution : Direction cosine : A line makes an angle α, β, γ with the positive direction of x, y

and z axes. Then $\cos \alpha, \cos \beta, \cos \gamma$ called direction cosine. It is denoted by l, m, n.

2nd part : We have

$$l + m + n = 0 \quad \dots \quad (1)$$

$$l^2 + m^2 - n^2 = 0 \quad \dots \quad (2)$$

$$\text{From (1) } n = -(l + m)$$

$$\text{Putting the values of } n \text{ in (2) We get}$$

$$l^2 + m^2 - (l + m)^2 = 0$$

$$\Rightarrow l^2 + m^2 - l^2 - 2lm - m^2 = 0$$

$$\Rightarrow -2lm = 0$$

$$\Rightarrow l = 0 \quad \dots \quad (3)$$

$$\text{or } m = 0 \quad \dots \quad (4)$$

$$\text{From (1) and (3)}$$

$$l + m + n = 0$$

$$l + 0m + 0n = 0$$

$$\therefore \frac{l}{0-0} = \frac{m}{1-0} = \frac{n}{0-1}$$

$$\therefore \frac{l}{0} = \frac{m}{1} = \frac{n}{-1} = \frac{1}{\sqrt{0^2 + 1^2 + 1^2}} = \frac{1}{\sqrt{2}}$$

$$\therefore l = 0, m = \frac{1}{\sqrt{2}}, n = -\frac{1}{\sqrt{2}}$$

$$\text{Again, from (1) and (4) We get,}$$

$$l + m + n = 0$$

$$0l + m + 0n = 0$$

$$\therefore \frac{l}{0-1} = \frac{m}{0-0} = \frac{n}{1-0}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{0} = \frac{n}{1} = \frac{1}{\sqrt{1^2 + 0^2 + 1^2}} = \frac{1}{\sqrt{2}}$$

$$\therefore l = \frac{-1}{\sqrt{2}}, m = 0, n = \frac{1}{\sqrt{2}}$$

$$l_1 l_2 + n_1 n_2 + m_1 m_2 = 0$$

$$\text{Now, } -\frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{-1}{3} + \frac{2}{3} \cdot \frac{2}{3} = -\frac{2}{9} - \frac{2}{9} + \frac{4}{9} = 0$$

Hence The given lines are perpendicular each other

17 Prove that $l^2 + m^2 + n^2 = 1$ where l, m, n are direction cosines of any line [(2000) N.U.H. 2002, 2008]

Solution. Let $P(x, y, z)$ be the co-ordinates of any point on the line

$$\therefore \cos \alpha = \frac{ON}{OP} = \frac{x}{r}$$

$$\therefore x = r \cos \alpha = rl$$

Similarly,

$$y = rm$$

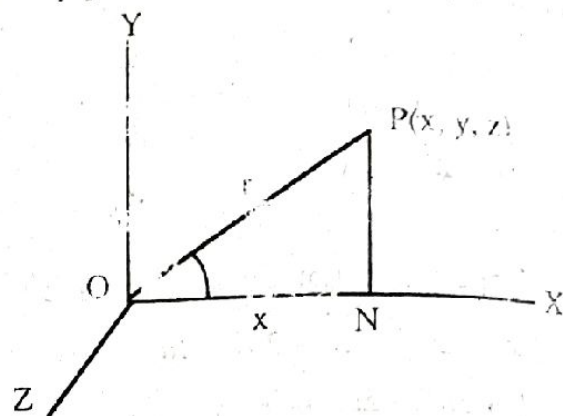
$$z = rn$$

$$\text{Also } OP^2 = r^2 = x^2 + y^2 + z^2$$

$$\text{Now, } x^2 + y^2 + z^2 = r^2 l^2 + r^2 m^2 + r^2 n^2$$

$$\Rightarrow r^2 = r^2 (l^2 + m^2 + n^2)$$

$$\therefore l^2 + m^2 + n^2 = 1 \text{ Proved}$$



Find the angle between two diagonals of a cube.

Solution.

Let OACNBP be a cube whose length of each edge is a . Let OP and CN be the diagonals of a cube.

$$\therefore \text{Co-ordinate of } O(0, 0, 0)$$

$$P(a, a, a)$$

$$N(a, a, 0)$$

$$C(0, 0, a)$$

$$\text{Dir's of } OP; a-0, a-0, a-0$$

$$\text{ie, } a, a, a$$

$$\text{Dir's of } CN; a-0, a-0, 0-a$$

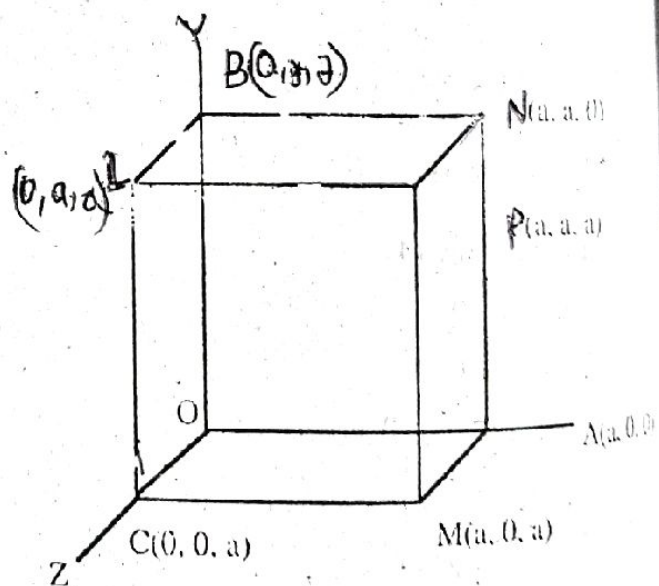
$$\text{ie, } a, a, -a$$

Let θ be the angle between two diagonals

Opened CN ,

$$\therefore \cos \theta = \frac{a \cdot a + a \cdot a + a \cdot (-a)}{\sqrt{(a^2 + a^2 + a^2)} \sqrt{(a^2 + a^2 + a^2)}} = \frac{a^2}{\sqrt{3a^2} \times \sqrt{3a^2}}$$

$$\cos \theta = \frac{1}{3} \quad \theta = \cos^{-1} \frac{1}{3} \text{ (Ans)}$$



$$\begin{aligned} \Rightarrow af + bg - ch &= \pm 2 \sqrt{af} \cdot \sqrt{bg} \\ \Rightarrow (\sqrt{af})^2 \pm 2 \sqrt{af} \cdot \sqrt{bg} + (\sqrt{bg})^2 &= (\sqrt{ch})^2 \\ \Rightarrow (\sqrt{af} \pm \sqrt{bg})^2 &= (\sqrt{ch})^2 \\ \Rightarrow \sqrt{af} \pm \sqrt{bg} &= \pm \sqrt{ch} \\ \Rightarrow \sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} &= 0 \text{ (proved)} \end{aligned}$$

19. If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of any two lines AB and CD and θ be the angle between them. Then prove that $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$.

Proof:

Let l_1, m_1, n_1 and l_2, m_2, n_2 be the dc's of AB and CD. Let OA = r_1 and OC = r_2 be the distances between AB and CD.

From the ΔOAC ;

$$\cos\theta = \frac{OA^2 + OC^2 - AC^2}{2 \cdot OA \cdot OC}$$

$$\Rightarrow 2r_1r_2 \cos\theta = r_1^2 + r_2^2 - \{(l_1r_1 - l_2r_2)^2 + (m_1r_1 - m_2r_2)^2 + (n_1r_1 - n_2r_2)^2\}$$

$$\Rightarrow 2r_1r_2 \cos\theta = r_1^2 + r_2^2 - l_1^2r_1^2 - 2r_1r_2l_1l_2 - l_2^2r_2^2 - m_1^2r_1^2 + 2r_1r_2m_1m_2 - m_2^2r_2^2 - n_1^2r_1^2 + 2r_1r_2n_1n_2 - n_2^2r_2^2$$

$$\Rightarrow 2r_1r_2 \cos\theta = r_1^2 + r_2^2 - r_1^2(l_1^2 + m_1^2 + n_1^2) - r_2^2(l_2^2 + m_2^2 + n_2^2) + 2r_1r_2(l_1l_2 + m_1m_2 + n_1n_2)$$

$$\Rightarrow 2r_1r_2 \cos\theta = r_1^2 + r_2^2 - r_1^2 - r_2^2 + 2r_1r_2(l_1l_2 + m_1m_2 + n_1n_2)$$

$$\therefore \cos\theta = l_1l_2 + m_1m_2 + n_1n_2 \text{ (Proved)}$$

20. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$.

Solution:

From the figure, eight vertices can be taken as Co-ordinate of O(0, 0, 0)

$$P(a, a, a)$$

$$N(a, a, 0)$$

$$C(0, 0, a)$$

$$A(a, 0, 0)$$

$$L(0, a, a)$$

$$M(a, 0, a)$$

$$B(a, 0, 0)$$

OP, CN, AL and BM are four diagonals of a Cube OANBLCMP

Dr's of OP $a=0, a=0, a=0$

ie, a, a, a

ie, 1, 1, 1

Let l, m, n be the de's of op

$$\frac{l}{1} = \frac{m}{1} = \frac{n}{1} = \frac{1}{\sqrt{1^2+1^2+1^2}} = \frac{1}{\sqrt{3}}$$

$$l = \frac{1}{\sqrt{3}}, m = \frac{1}{\sqrt{3}}, n = \frac{1}{\sqrt{3}}$$

$$\text{ii De's of op } \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$$

$$\text{Similarly, de's of CN; } \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$$

$$\text{de's of AL; } \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$$

$$\text{de's of BM; } \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$$

Let l, m, n be The de's of the line which makes an angle $\alpha, \beta, \gamma, \delta$ with the diagonals op, CN, AL and BM.

$$\therefore \cos \alpha = l \cdot \frac{1}{\sqrt{3}} + m \cdot \frac{1}{\sqrt{3}} + n \cdot \frac{1}{\sqrt{3}}$$

$$\therefore \cos \alpha = \frac{l+m+n}{\sqrt{3}}$$

$$\text{Similarly, } \cos \beta = \frac{l+m-n}{\sqrt{3}}$$

$$\cos \gamma = \frac{-l+m+n}{\sqrt{3}}$$

$$\cos \delta = \frac{l-m+n}{\sqrt{3}}$$

$$\text{Now, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta$$

$$= \left(\frac{l+m+n}{\sqrt{3}} \right)^2 + \left(\frac{l+m-n}{\sqrt{3}} \right)^2 + \left(\frac{-l+m+n}{\sqrt{3}} \right)^2 + \left(\frac{l-m+n}{\sqrt{3}} \right)^2$$

$$= \frac{1}{3} [l^2 + m^2 + n^2 + 2lm + 2mn + 2ln + 2lm - 2mn - 2nl + l^2 + m^2 + n^2 + l^2 + m^2 + n^2 - 2lm + 2mn - 2nl + l^2 + m^2 + n^2 - 2lm - 2mn + 2nl]$$

$$= \frac{1}{3} [1+1+1+1] = 4/3$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3 \text{ (Proved)}$$

EXERCISE-1

1. Find the distance between the origin and point given.

(a) (3, 4, 5)

Ans. 5

(b) (-4, 0, -2)

Ans. 2

2. Find the distance between each of the following pairs of points.

(a) (5, -2, 3), (-4, 5, 7)

Ans. 10

(b) (3, 3, 0), (6, 0, 2)

Ans. 5

3. Find the co-ordinates of the point which divides the line joining the points (2, -4, 3) and (-4, 5, -6) in the ratio 1:2.

Ans. (4, -7, 1)

4. Find the ratio in which the line joining the points (2, 4, 5) and (3, 5, -4) is divided by the YZ plane.

Ans. -2 or 2:1

5. If P and Q are (2, 3, -6) and (3, -4, 5) respectively and O be the origin find the direction cosines of OP, OQ, PQ.

[R.U. 1980]

Ans. $2/7, 3/7, -6/7; 3/5\sqrt{2}, -4/5\sqrt{2}, 1/\sqrt{2}; (-1/7, -11/7, -17/7)$

6. Find the angle between the lines whose direction ratios are (3, 1, 2) and (2, -2, 4).
Ans. $\cos^{-1} \sqrt{3}/7$

7. Find the direction cosines of the positive z-axis.

Ans. 0, 0, 1

8. Find the cosine of the angle between two directed lines having the direction cosines

(a) $4/9, -8/9, -1/9; -1/3, -2/3, -2/3$.

Ans. $\cos^{-1} \sqrt{14}/7$

(b) $1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}; 1/2, \sqrt{6}/4, -1/2$

Ans. $\cos^{-1} 1/2$

9. Show that the lines whose direction cosines are proportional to 2, 1, 1; 4, $\sqrt{3}-1, -\sqrt{3}-1$; $4, -\sqrt{3}-1, \sqrt{3}-1$, are inclined to each other at angle $\pi/3$.

10. If P, Q are $(x_1, y_1, z_1), (x_2, y_2, z_2)$, the projection of PQ on a line whose direction cosines are $\cos \alpha, \cos \beta, \cos \gamma$ is given by $(x_2 - x_1) \cos \alpha + (y_2 - y_1) \cos \beta + (z_2 - z_1) \cos \gamma$

11. Find the distance of (-1, 2, 5) from the line through (3, 4, 5) whose direction cosines are proportional to 2, -3, 6.

Ans. $4\sqrt{61}/13$

12. (a) Find the distance of (-2, 3, 4) from the line through the point (-1, 3, 2) whose direction cosines are proportional to 12, 3, -4.

Ans. $\sqrt{445}/13$ [D.U. 1980]

12. (b) The direction cosines l, m, n of two lines are connected by the relations $l + m + n = 0$ and $2lm + 2ln - mn = 0$. Find them.

Ans. $1/\sqrt{6}, 1/\sqrt{6}, -2/\sqrt{6}; 1/6, -2/\sqrt{6}, 1/\sqrt{6}$.

13. The direction cosines of two lines are determined by the relations

Art. 18. (a) Definition of a plane : A plane is a surface such that if any two points are taken on it, the straight line joining them lies wholly and the surface i, e. every point of the line joining the two point will be on the plane.

Art. 18. (b) : To prove that the general equation of the first degree in x, y, z , i.e. $ax + by + cz + d = 0$ represents a plane.

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be the co-ordinates of any two points lying on the locus whose equation is

$$ax + by + cz + d = 0$$

$$\therefore ax_1 + by_1 + cz_1 + d = 0 \quad \dots \dots \dots (1)$$

$$\text{and } ax_2 + by_2 + cz_2 + d = 0 \quad \dots \dots \dots (2)$$

$$\text{Multiplying (3) by } k \text{ and adding to (2), we have} \quad \dots \dots \dots (3)$$

$$a(x_1 + kx_2) + b(y_1 + ky_2) + c(z_1 + kz_2) + d(1 + k) = 0$$

$$\text{Or, } \frac{a(x_1 + kx_2)}{1 + k} + \frac{b(y_1 + ky_2)}{1 + k} + \frac{c(z_1 + kz_2)}{1 + k} + d = 0 \quad \dots \dots \dots (4)$$

Result (4) shows that equation (1) is satisfied by the point whose co-ordinates are

$$\left(\frac{x_1 + kx_2}{1 + k}, \frac{y_1 + ky_2}{1 + k}, \frac{z_1 + kz_2}{1 + k} \right)$$

But from equation (6). Art. 4, it represents any point on the line joining (x_1, y_1, z_1) , (x_2, y_2, z_2) ; and giving different value to k . We get all the points on the line. Therefore the line completely lies on the locus; in other words, the locus is a plane.

Conclusion : Hence every equation of the first degree in x, y, z represents a plane.

Cor. 1. General equation of a given plane that passes through a given point.

$$\text{The general equation of a plane is } ax + by + cz + d = 0 \quad \dots \dots \dots (5)$$

Since it passes through the point (x_1, y_1, z_1) , we have

$$ax_1 + by_1 + cz_1 + d = 0 \quad \dots \dots \dots (6)$$

Subtracting (6) from (5), we have

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0 \quad \dots \dots \dots (28)$$

Cor. 2. General equation of a plane through the origin.

In case the plane passes through origin $(0, 0, 0)$ the general equation of the plane $ax + by + cz + d = 0$ will be satisfied by $(0, 0, 0)$ so that $d = 0$. Hence the equation reduces to the form.

$$ax + by + cz = 0 \quad \dots \dots \dots (29)$$

Cor. 3. The equation of a plane passing through the two points

(x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$L = (x - x_1)(y - y_2) - (x - x_2)(y - y_1) + R[(x - x_1)(z - z_2)$$

$$(x - x_2)(z - z_1) - \dots = 0$$

It may easily be verified that 29 (a) is the equation of a plane passing through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) because this equation is satisfied by the co-ordinates for all values of x and further in 29 (a) the terms xy cancel out, so that 29(a) is an equation of the first degree in x and z and as such it is of a plane.

The equation may also be written as

$$(x - x_1)(y - y_2) - (x - x_2)(y - y_1)$$

$$= A[(x - x_1)(z - z_2) - (x - x_2)(z - z_1)]$$

Art. 19. Plane through three given points. To find the equation of the plane passing through the three non-collinear points.

$(x_1, y_1, z_1); (x_2, y_2, z_2), (x_3, y_3, z_3)$

Let the equation to the plane be

$$ax + by + cz + d = 0$$

As the given points lie on the plane (1), we have

$$ax_1 + by_1 + cz_1 + d = 0$$

$$ax_2 + by_2 + cz_2 + d = 0$$

$$ax_3 + by_3 + cz_3 + d = 0$$

Eliminating a, b, c, d from (1), (2), (3) and (4) we have

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \text{ or, } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \dots \quad (20)$$

which is the required equation of the plane.

Art. 20. Standard forms of the equation of a plane.

(a) **Intercept form :** To find the equation to the plane in terms of the intercepts, which it makes on the axes.

Let the intercepts made by the plane on the axes be a, b, c . Then we can regard it as a plane passing through the points $(a, 0, 0); (0, b, 0); (0, 0, c)$. Then by Art. 19 its equation is

$$\begin{vmatrix} x & y & z & 1 \\ a & 0 & 0 & 1 \\ 0 & b & 0 & 1 \\ 0 & 0 & c & 1 \end{vmatrix} = 0$$

or, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (31)

which is the required equation.

Putting the values of l, m, n , and p in $lx + my + nz = p$ we have

$$\frac{a}{\sqrt{(a^2 + b^2 + c^2)}}x + \frac{b}{\sqrt{(a^2 + b^2 + c^2)}}y + \frac{c}{\sqrt{(a^2 + b^2 + c^2)}}z = -\frac{d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ if } d \text{ is negative}$$

Again if d is positive, then from (iv) we must choose

$$p = \frac{d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ and hence from (3) } l = \frac{-a}{\sqrt{(a^2 + b^2 + c^2)}}$$

$$m = \frac{-b}{\sqrt{(a^2 + b^2 + c^2)}}; \quad n = \frac{-c}{\sqrt{(a^2 + b^2 + c^2)}}$$

Put the values of l, m, n , and p in (ii), then

$$\frac{-a}{\sqrt{(a^2 + b^2 + c^2)}}x - \frac{b}{\sqrt{(a^2 + b^2 + c^2)}}y - \frac{c}{\sqrt{(a^2 + b^2 + c^2)}}z = \frac{d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ if } d \text{ is positive}$$

Thus the normal form of the equation $ax + by + cz + d = 0$ is

$$\left. \begin{aligned} -\frac{a}{\sqrt{\sum a^2}}x - \frac{b}{\sqrt{\sum a^2}}y - \frac{c}{\sqrt{\sum a^2}}z &= \frac{d}{\sqrt{\sum a^2}} \text{ if } d \text{ is positive} \\ \frac{a}{\sqrt{\sum a^2}}x + \frac{b}{\sqrt{\sum a^2}}y + \frac{c}{\sqrt{\sum a^2}}z &= \frac{d}{\sqrt{\sum a^2}}; \text{ if } d \text{ is negative} \end{aligned} \right\} \dots \dots \dots (32)$$

Note : Since $\frac{a}{\sqrt{\sum a^2}}, \frac{b}{\sqrt{\sum a^2}}, \frac{c}{\sqrt{\sum a^2}}$ are actual direction cosines of the normal to the plane $ax + by + cz + d = 0$, therefore a, b, c are proportional to the d.c.'s of the normal to the plane.

Art. 22. (a) : Angle between two planes "Angle between two planes is equal to the angle between the normals to them from any point. Thus the angle between the two planes.

$$ax + by + cz + d = 0$$

$$\text{and } a_1x + b_1y + c_1z + d = 0$$

is equal to the angle between the lines with direction ratios a, b, c and a_1, b_1, c_1 . Hence if θ be the angle between the normals, then

$$\cos \theta = \frac{aa_1 + bb_1 + cc_1}{\sqrt{(a^2 + b^2 + c^2)} \sqrt{(a_1^2 + b_1^2 + c_1^2)}}$$

Art. 22. (b) : Condition of perpendicularity of two planes.

If two planes are perpendicular to each other, so are their normals whose direction cosines are proportional to a_1, b_1, c_1 , and a_2, b_2, c_2 . Hence the condition of perpendicularity is

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

Art. 22. (c) : Condition of parallelism of two planes.

Two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are parallel if and only if the co-efficients of x, y, z in their equations are proportional, i.e., if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Art. 23. Some system of planes.

(i) Plane parallel to a given plane and passing through a given point.

The equation of a plane parallel to a given plane $ax + by + cz + d = 0$ is of the form

$$ax + by + cz + k = 0 \quad \dots \dots \dots (1) \quad \dots \dots \dots (35)$$

EXAMPLES

Ex. 1. Find the equation of the plane through the points (2, 3, 1), (1, 1, 3) and (2, 2, 3) also the perpendicular distance from the point (5, 6, 7) to this plane.

First Method : The required plane is

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{vmatrix}$$

or, $2x - 2y - z + 3 = 0$

2nd Method :

The equation of any plane through (2, 3, 1) is

$$a(x-2) + b(y-3) + c(z-1) = 0 \quad \dots \quad (1)$$

Since it passes through (1, 1, 3) and (2, 2, 3) we have

$$a(1-2) + b(1-3) + c(3-1) = 0 \text{ or, } a + 2b - 2c = 0 \quad \dots \quad (2)$$

$$\text{and } a(2-2) + b(2-3) + c(3-1) = 0 \text{ or, } b - 2c = 0 \quad \dots \quad (3)$$

Solving (2) and (3) for a, b, c by cross multiplication, we have $\frac{a}{-2} = \frac{b}{2} = \frac{c}{1}$

Putting these value in (1) we have

$$-2(x-2) + 2(y-3) + 1(z-1) = 0 \text{ or, } 2x - 2y - z + 3 = 0$$

The perpendicular distance P is from (5, 6, 7) to the (4) is

$$P = \frac{2.5 - 2.6 - 7 + 3}{\sqrt{4 + 4 + 1}} = 2$$

3rd Method.

See Art. 18. (a) cor. 3

The equation of plane passing through (2, 3, 1) and (1, 1, 3) is by Art. 18 (a) cor. 3, eq. 29

$$(x-2)(y-1) - (x-1)(y-3) = A\{(x-2)(z-3) - (x-1)(z-1)\},$$

Since it passes through (2, 2, 3), hence

$$0 - (2-1)(2-3) = A\{0(2-1) - (3-1)\} \text{ or, } A = -1/2$$

Now putting the value of A, we have

$$(x-2)(y-1) - (x-1)(y-3) = \frac{1}{2} \{(x-2)(z-3) - (x-1)(z-1)\}$$

$$\text{or } 2x - 2y - z + 3 = 0$$

Ex. 2. Show that the four points (0, -1, -1), (4, 5, 1), (3, 9, 4) and (-4, 4, 4) lie on a plane

The four points are coplanar if the determinant

$$\begin{vmatrix} 0 & -1 & -1 & 1 \\ 4 & 5 & 1 & 1 \\ 3 & 9 & 4 & 1 \\ -4 & 4 & 4 & 1 \end{vmatrix} \text{ is zero}$$

Otherwise we shall find the equation of the plane passing through any three points say $(0, -1, -1)$, $(4, 5, 1)$ and $(3, 9, 4)$ and then show that the fourth point $(-4, 4, 4)$ satisfied the plane.

Equations of any plane through $(0, -1, -1)$, $(4, 5, 1)$ and $(3, 9, 4)$ is

$$5x - 7y + 11z + 4 = 0 \text{ by example (1)}$$

The point $(-4, 4, 4)$ clearly satisfies this plane. Hence the four points are coplanar.

N.B : For finding out the equation to the plane use, Cor, 3

Ex. 3. Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and the point $(1, 2, 3)$

Any plane through the intersection of the two planes is

$$x + 2y + 3z + 4 + k(4x + 3y + 2z + 1) = 0 \quad \dots \quad \dots \quad (1)$$

Since it passes through $(1, 2, 3)$,

$$18 + 17k = 0 \quad \text{or } k = -18/17$$

Putting the value of k in (1) we have

$$x + 2y + 3z + 4 - \left(\frac{18}{17}\right)(4x + 3y + 2z + 1) = 0$$

$$\text{or, } 55x + 20y - 15z - 50 = 0 \text{ or, } 11x + 4y - 3z = 10$$

which is the required plane.

Ex. 4. Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$

Let θ be the angle between the planes then

$$\cos \theta = \frac{2 \cdot 1 + (-1) \cdot 1 + 1 \cdot 2}{\sqrt{2^2 + (-1)^2 + 1^2} \sqrt{1^2 + 1^2 + 2^2}} = \frac{3}{\sqrt{6}\sqrt{6}} = \frac{1}{2} \therefore \theta = \frac{\pi}{3}$$

Ex. 5. Find the equation of the plane through the point $(4, -2, 1)$ and parallel to the plane whose direction numbers are $7, 2, -3$.

The equation to the plane through $(4, -2, 1)$ is $a(x - 4) + b(y + 2) + c(z - 1) = 0$. The condition that the co-efficients a, b, c are proportional to the direction numbers $7, 2, -3$, therefore, the equation of the required plane is $7k(x - 4) + 2k(y + 2) - 3k(z - 1) = 0$ where k is constant.

$$\text{or, } 7(x - 4) + 2(y + 2) - 3(z - 1) = 0 \text{ or, } 7x + 2y - 3z - 21 = 0$$

Ex. 6. Find the equation of the plane through the point $(4, 0, 1)$ and parallel to the plane $4x + 3y - 12z + 6 = 0$

The equations of the parallel planes differ by a constant only. Hence the equation of the plane parallel to $2x + 3y - 12z + 6 = 0$ is $4x + 4y - 12z + k = 0$ Since it passes through $(4, 0, 1)$.

$$16 + 0 - 12 + k = 0. \text{ or, } k = -4$$

$$\text{The required plane is } 4x - 3y - 12z - 4 = 0$$

Ex. 7. Find the equation of the plane passing through the lines of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z + 1 = 0$

$$\text{The plane } 2x - y + k(3z - y) = 0$$

$$\text{i.e. } 2x - (1 + k)y + 3kz = 0$$

passes through the intersection of the two given planes whatever be the value of k .

It will be perpendicular to $4x + 5y - 3z + 1 = 0$

$$\therefore 2 \cdot 4 - (1+k)5 + 3k(-3) = 0 \text{ i.e. } 14k = 3 \text{ or, } k = 3/14$$

Thus the required plane is $2x - y + 3(3z - y)/14 = 0$

$$\text{or, } 28x - 17y + 9z = 0$$

Ex. 8. Find the equation of the plane through the pts. (2, 2, 1) and (9, 3, 6) perpendicular to the plane $2x + 6y + 6z = 9$

[C.H. 1980; C.U. 1981; R.U. 1962, '78; D.H. 1976] D.U.(P) 1982

Any plane through (2, 2, 1) is

$$a(x-2) + b(y-2) + c(z-1) = 0 \quad \dots \dots \dots (1)$$

Since it passes through (9, 3, 6) then (1) becomes

$$a(9-2) + b(3-2) + c(6-1) = 0 \text{ or, } 7a + b + 5c = 0 \quad \dots \dots \dots (2)$$

The plane (1) is perpendicular to $2x + 6y + 6z - 9 = 0$

$$\text{Hence } 2a + 6b + 6c = 0 \quad \dots \dots \dots (3)$$

From (2) and (3) by cross multiplication.

$$\frac{a}{-24} = \frac{b}{-32} = \frac{c}{40} \text{ or, } \frac{a}{3} = \frac{b}{4} = \frac{c}{-5} \quad \dots \dots \dots (4)$$

Putting the values of a, b, c , from (4) in (1), the equation of the required plane is $3(x-2) + 4(y-2) - 5(z-1) = 0$ or, $3x + 4y - 5z = 9$

Alternate Method : The required plane is

$$\begin{vmatrix} x-2 & y-2 & z-1 \\ 9-2 & 3-2 & 6-1 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

$$\text{or, } 3x + 4y - 5z = 9 \quad \text{by Art. (11)}$$

Ex. 9. Show that the equation to the plane through P (2, 3, -1) at right angles to OP is $2x + 3y - z = 14$

The direction cosines of the line OP are $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}$

Any plane at right angles to OP can be written as

$$\frac{2}{\sqrt{14}}x + \frac{3}{\sqrt{14}}y - \frac{1}{\sqrt{14}}z = p$$

$$\text{Where } p = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{14}$$

$$\text{The required plane is } 2x + 3y - z = 14$$

Ex. 10. A plane meets the Co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r); show that the equation of the plane is

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$$

$$\text{Let the equation of the plane be } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \dots \dots \dots (1)$$

$\therefore$ A is (a, 0, 0) B is (0, b, 0) and C is (0, 0, c)

$$\therefore a(3-1)b(-2-2) + c(6+2) = 0$$

$$\Rightarrow 2a - 4b + 8c = 0$$

$$\Rightarrow a - 2b + 4c = 0 \quad \dots \quad (3)$$

According to the problem, planes (1) and (2) are perpendicular.

$$\therefore 2a - b - c = 0 \quad \dots \quad (4)$$

From (3) and (4), by the rule of cross multiplication

$$\frac{a}{2+4} = \frac{b}{8+1} = \frac{c}{-4+1}$$

$$\frac{a}{6} = \frac{b}{9} = \frac{c}{-3} = k \text{ (say)}$$

$$\therefore a = 6k, b = 9k, c = -3k$$

Putting the values of a, b, c in (2) we get, $6x(x-1) + 9k(y-2) - 3k(z+2) = 0$

$$2x + 3y - z - 10 = 0$$

Which is required equation of the plane

EXERCISE II

1. Find the general equations of planes perpendicular to

(1) x -axis (2) y -axis (3) xy plane (4) xoz plane.

Ans. (1) $x = p$; (2) $y = p$; (3) $lx + my = p$; (4) $lx + nz = p$

2. Find the equation of the plane through the points.

(a) $(2, 1, 3), (-1, -2, 4), (4, 2, 1)$

$$\text{Ans. } 5x - 4y + 3z - 15 = 0$$

(b) $(1, 1, -1), (-2, -2, 2), (-1, -1, 2)$

$$\text{Ans. } x - 3y - 2z = 0$$

(c) $(2, 1, -3), (3, -1, 4), (7, 5, 6)$

$$\text{Ans. } 23z - 13y - 7x - 54 = 0$$

3. Show that the four points are coplanar

(a) $(0, 4, 3), (-1, -5, -3), (-2, -2, 1), (1, 1, -1)$

(b) $(-6, 3, 2), (3, -2, 4), (5, 7, 3)$ and $(-13, 17, -1)$

4. Show that the line of intersection of the first two planes is co-planar with the line of intersection of the latter two and find the equation of the plane containing two lines.

(a) $x + 2y - z - 3 = 0, 3x - y + 2z - 1 = 0,$

$$2x - 2y + 3z - 2 = 0, x - y + z + 1 = 0$$

$$\text{Ans. } 7x - 7y + 8z + 3 = 0$$

(b) $7x - 4y + 7z + 16 = 0, 4x + 3y - 2z + 3 = 0$

$$x - 3y + 4z + 6 = 0, x - y + z + 1 = 0$$

$$\text{Ans. } 3x - 7y - 9z + 13 = 0$$

5. Find the smallest angle between the planes

(a) $3x + 2y - 5z - 4 = 0, 2x - 3y + 5z - 8 = 0$

(b) $3x - 4y + 5z = 0, 2x - y - 2z = 0$

(c) $2x + 2y - 7 = 0, x + 2y - z + 16 = 0$

$$\text{Ans. } \cos \theta = \frac{25}{38}$$

$$\text{Ans. } \pi/2$$

$$\text{Ans. } 30^\circ$$

6. Find direction cosines of the normals to the planes and length of the normal.

(a) $x + 2y + 2z - 1 = 0$

Ans. $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}, P = \frac{1}{3}$

(b) $9x + 6y - 2z + 7 = 0$

Ans. $-\frac{9}{11}, -\frac{6}{11}, \frac{2}{11}, P = \frac{7}{11}$

7. Find the intercepts made by the plane $2x + y - 2z = 3$ on the co-ordinate axes.

What are the d.c.s of the plane.

Ans. $\frac{3}{2}, 3, -\frac{3}{2}$; d.c.s. $(\frac{2}{3}, \frac{1}{3}, -\frac{3}{2})$

8. A plane contains the pt. $(1, -1, -2)$. Its normal has d.r's $3, -2, -1$. Find the equation of the plane.

Ans. $3x - 2y - z = 7$

9. Find the equations to the planes

(a) Through the intersections of the planes $x - 2y + 3z + 4 = 0$ and $2x - 3y + 4z - 7 = 0$ and the point $(1, -1, 1)$

Ans. $9x - 13y + 17z - 39 = 0$

(b) Through $(1, 1, 2)$ and perpendicular to each of the planes $2x - 2y - 4z - 6 = 0$ and $3x + y + 6z - 4 = 0$

Ans. $x + 3y - z - 2 = 0$

(c) Through $(1, 2, 2)$ and parallel to $3x + 2y + z = 7$

Ans. $3x + 2y + z - 9 = 0$

(d) through $(-1, 3, 2)$ and perpendicular to the two planes

$x + 3y + 2z = 5, 3x + 3y + 2z = 8$

Ans. $2x + 4y + 3z + 8 = 0$

10. Find the equation of the plane which is parallel to the plane

(a) $4x - 4y + 7z - 3 = 0$ and distance 4 units from the point $(3, 1, -2)$.

Ans. $4x - 4y + 7z + 38 = 0, 4x - 4y + 7z - 34 = 0$

(b) $2x - 3y - 6z - 14 = 0$ and distant 5 from the origin.

Ans. $2x - 3y - 6z \pm 35 = 0$

11. Find the equation of the plane which passes through the points $(1, 0, -1)$ and $(2, 1, 3)$ and is perpendicular to the plane $2x + y + z = 1$

Ans. $3x - 7y + z = 2$

[N.U.H. 2005] [C. U; D. U. 1977]

(a) Find the equation of the plane perpendicular to each of the $x - 4y + z = 0$ and $3x + 4y + z - 2 = 0$ and at a distance unity from the origin.

Ans. $-4x + y + 8z + 9 = 0$ [D.U. 1982]

12. Plane is drawn through the line of intersection of the planes $x + 2y + 2z = 1, x + y - z + 1 = 0$ and is at distances. 1 (one) from the point $(4, -2, 1)$. Prove that there are two such planes and find their equations.

Ans. $3x + 4y + 1 = 0, 3x + 2y - 6z + 5 = 0$ [R.U. 63]

13. Find the equation of the plane which is perpendicular to the plane $5x + 3y + 6z + 8 = 0$ and which contains the line of intersection of the planes $x + 2y + 3z - 4 = 0$, and $2x + y - z + 5 = 0$

Ans. $51x + 15y - 50z + 173 = 0$ [C.U. '80; C.H. '80]

14. Find the equation of plane that passes through $(2, -3, 1)$ and is normal to the line joining the points $(3, 4, -1)$ and $(2, -1, 5)$

Ans. $x + 5y - 6z + 19 = 0$ [D.U. '82; C.U. '78]

15. Find the equation of the plane through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and perpendicular to the plane $x + y + z + 9 = 0$ and show that it is perpendicular to xz plane. Ans. $x - z = 2$, the Co-efficient of y is zero.

THE STRAIGHT LINE

3

Art. 28. General equation of a straight line.

A line be determined as the intersection of any two planes through it.

No if we take two equations of the first degree in x, y and z , together, i. e. $a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$... (1)

then any point which satisfies these equations individually will satisfy both of them as well. Hence these two equations taken together represent the line of intersection of the two given planes and we know that two planes intersect in a straight line.

Then the equation (1) represents the equation of a straight line which is the line of intersection of the two planes.

Conclusion : Thus a straight line in space as represented by two equations of the first degree in x, y, z .

Art. 29. Parametric equations of the line

Let the direction cosines of the line L be l, m, n and let $P_1(x_1, y_1, z_1)$ be a point on the line. Then L is the locus of $P(x, y, z)$ moving so that

$$x - x_1 = r \cos \alpha = rl$$

$$y - y_1 = r \cos \beta = rm$$

$$z - z_1 = r \cos \gamma = rn$$

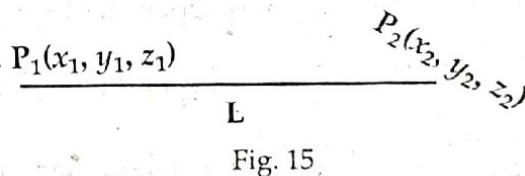


Fig. 15

$$\therefore x = x_1 + rl, y = y_1 + rm \text{ and } z = z_1 + rn \quad \dots \quad (43)$$

where the parameter represents the variable length P_1P .

If a, b, c , be direction ratios of L these equations can be written as $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$... (44)

Here t does not represent the distance of the variable length P_1P .

The equation represents the co-ordinates of any point on the line L by giving different values to ' t '.

Art. 30. (a) Symmetrical Form : To find the equation of a straight line through a given point (x_1, y_1, z_1) in a given direction.

Let the given point be $P_1(x_1, y_1, z_1)$ and let the given direction be defined by the direction cosines l, m, n .

Let $P(x, y, z)$ be a point on the line and let the measure of P_1P be r .

Then projecting P_1P on the co-ordinate axes, we have

$$x - x_1 = lr, y - y_1 = mr, z - z_1 = nr$$

So that for all points (x, y, z) on the given line.

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = t$$

Thus $\boxed{\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}}$

is the symmetrical form of the equation of a straight line

(b) Symmetrical form of direction ratios.

If a, b, c represent the direction ratios then the direction cosines are

$$\frac{a}{\sqrt{(\sum a^2)}}, \frac{b}{\sqrt{(\sum a^2)}}, \frac{c}{\sqrt{(\sum a^2)}}$$

Thus the equations of the line (1) take the form

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = r$$

$$\frac{a}{\sqrt{(\sum a^2)}}, \frac{b}{\sqrt{(\sum a^2)}}, \frac{c}{\sqrt{(\sum a^2)}}$$

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = t$$

Here t does not represent the actual length of any point on the line from (x_1, y_1, z_1) . t is a parameter.

(c) If this line is perpendicular to one of the co-ordinate axes the equation takes one of the following :

$$x = x_1, \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad (\text{perpendicular to } x\text{-axis})$$

$$y = y_1, \frac{x-x_1}{a} = \frac{z-z_1}{c} \quad (\text{perpendicular to } y\text{-axis})$$

$$z = z_1, \frac{x-x_1}{a} = \frac{y-y_1}{b} \quad (\text{perpendicular to } z\text{-axis})$$

(d) If this line perpendicular to two axes two equations are sufficient to determine the line.

$$x = x_1, y = y_1, \quad (\text{perpendicular to } x \text{ and } y\text{-axis})$$

$$x = x_1, z = z_1, \quad (\text{perpendicular to } x \text{ and } z\text{-axis})$$

$$y = y_1, z = z_1, \quad (\text{perpendicular to } y \text{ and } z\text{-axis})$$

Art. 32. Line through two given points.

If P, Q are $(x_1, y_1, z_1), (x_2, y_2, z_2)$ the direction ratios of PQ are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$ and therefore the equations to PQ are $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$... (40)

Art. 31. (a) Direction ratios from equation of a straight line.

Let the equations of the straight line be

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$$

Then the equation of a line which is parallel to (1) and passes through the origin are

$$a_1x + b_1y + c_1z + d_1 = 0 \quad a_2x + b_2y + c_2z + d_2 = 0$$

Solving equations $a_1x + b_1y + c_1z = 0$

and $a_2x + b_2y + c_2z = 0$ we have

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_2c_1 - a_1c_2} = \frac{z}{a_1b_2 - a_2b_1} \quad \dots \quad (51)$$

Which show the direction ratios of the line proportional to

$$b_1c_2 - b_2c_1, a_2c_1 - a_1c_2, a_1b_2 - a_2b_1 \quad \dots \quad (52)$$

(b) To transform the general form of the equation (1) of 32 (a) of a straight line to symmetrical form.

In order to find the co-ordinates of any point on the line we can choose it in a number of ways. For convenience let the point be chosen to be one where the line (1) of ways. For convenience let the point be chosen to be one where the line (1) of Art. 32 (a) cuts the xy plane i.e. where $z = 0$, putting $z = 0$ in (1) we have

$$a_1x + b_1y + d_1 = 0, a_2x + b_2y + d_2 = 0$$

$$\frac{x}{b_1d_2 - b_2d_1} = \frac{y}{d_1a_2 - a_1d_2} = \frac{1}{a_1b_2 - b_2a_1}$$

$\therefore$ Hence the co-ordinates of the required point are

$$\left(\frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1}, \frac{d_1a_2 - d_2a_1}{a_1b_2 - a_2b_1}, 0 \right) \quad \dots \quad (54)$$

In case $a_1b_2 - a_2b_1 = 0$, then instead of putting $z = 0$ we may put $x = 0$ or $y = 0$

Thus, in the symmetrical form, the equations of the given line whose direction ratios are proportional to $(b_1c_2 - b_2c_1), (a_2c_1 - a_1c_2), (a_1b_2 - a_2b_1)$ by eq. 4 of Art. 32 (a) and which passes through a point $(x_1, y_1, 0)$ by eq (5) are

$$\frac{x - x_1}{b_1c_2 - b_2c_1} = \frac{y - y_1}{a_2c_1 - a_1c_2} = \frac{z - 0}{a_1b_2 - a_2b_1} \quad \dots \quad (56)$$

[C. U. 1981]

Art. 33. Distance of a point from a line.

Let the length of the perpendicular distance from a point $P(x_1, y_1, z_1)$ to the line

$$\frac{x - a}{l} = \frac{y - b}{m} = \frac{z - c}{n} \quad \text{be } p \quad \dots \quad (1)$$

If A be the point (a, b, c) on the given line and N the foot of the perpendicular from P on it, we have

$$PN^2 = AP^2 - AN^2 \quad \dots \quad (2)$$

$$\text{But } AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$$

and $AN = \text{projection of } AP \text{ on the given line}$

$$= l(x_1 - a) + m(y_1 - b) + n(z_1 - c) \quad \text{provided } l, m, n$$

are the actual direction cosines of AP .

$$\therefore PN^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2 - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2$$

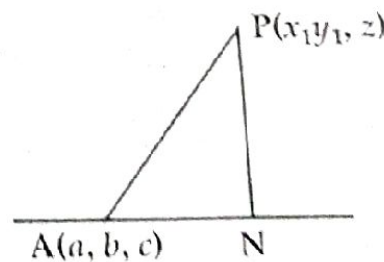


Fig. 16

find the condition of coplanarity.

Eliminate x, y, z from the above equations the eliminant is the required condition for coplanarity of the two lines this is

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = 0 \quad \dots \quad \dots$$

Definition :- When two lines do not intersect and are parallel as well i. e. they do not lie in the same plane, then these lines are said to be non-intersecting lines. The straight line which is perpendicular to each of these two non intersecting line is called the line of shortest distance and length of this line intercepted between the given lines is called the length of the shortest distance.

(b) To find the shortest distance between two given lines whose equation are given also obtain the equations of the shortest distance.

Let ABL and CDM be the two given lines whose equations are

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad \dots \quad \dots \quad (1)$$

$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad \dots \quad \dots \quad (2)$$

Where the points B and D are $(x_1, y_1, z_1), (x_2, y_2, z_2)$

Let LM be the line which

is perpendicular to both the given lines and let its direction cosines be l, m, n

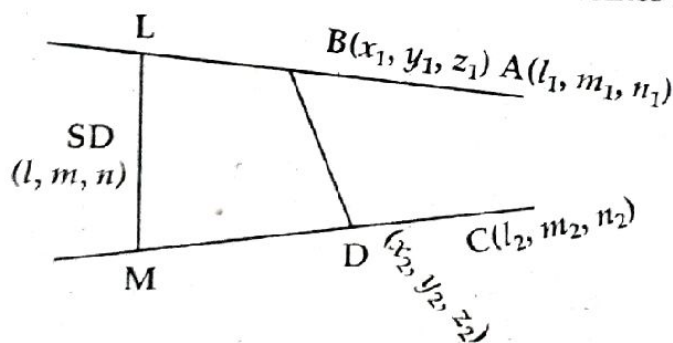


Fig: 17

Then by art 17. (c)

$$ll_1 + mm_1 + nn_1 = 0$$

$$ll_2 + mm_2 + nn_2 = 0$$

$$\therefore \frac{l}{m_1n_2 - m_2n_1} = \frac{m}{n_1l_2 - n_2l_1} = \frac{n}{l_1m_2 - l_2m_1}$$

(by cross multiplication)

$$\frac{\sqrt{(l^2 + m^2 + n^2)}}{\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}} = \frac{1}{\sin \theta} \quad \dots \quad (3) \quad \text{By Art. 17(a)}$$

$$\therefore l = \frac{m_1n_2 - m_2n_1}{\sin \theta}, m = \frac{n_1l_2 - n_2l_1}{\sin \theta}, n = \frac{l_1m_2 - l_2m_1}{\sin \theta} \quad \dots \quad (4)$$

The shortest distance (S. D) i. e. LM is the projection of BD on line LM whose direction cosines are l, m, n ,

$$S.D = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

$$= \frac{(x_2 - x_1)(m_1 n_2 - m_2 n_1) + (y_2 - y_1)(n_1 l_2 - n_2 l_1) + (z_2 - z_1)(l_1 m_2 - l_2 m_1)}{\sin \theta}$$

$$= \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} \div \sin \theta \text{ by (4)}$$

$$= \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} \div \sqrt{\Sigma(m_1 n_2 - m_2 n_1)^2}$$

[By (3) (6)]

Note : If the given lines are coplanar the S. D. is zero ;

Therefore, $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad \dots \quad \dots \quad \dots \quad (62)$

Art. 38. To find the equations the shortest distance we see that it is coplanar with both the given lines.

The equation of the plane containing the line (1) and the line of shortest distance whose direction cosines are l, m, n ,

is $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l & m & n \end{vmatrix} = 0 \quad \dots \quad \dots \quad \dots \quad (5) \dots \dots (63)$

Similarly the equation of the plane containing the line (2) and the shortest distance is

$$\begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ l & m & n \end{vmatrix} = 0 \quad \dots \quad \dots \quad \dots \quad (6) \dots \dots (64)$$

These two planes (2) and (2) taken together represent the equation of the line of shortest distance

N. B. There are other methods of determining the shortest distance. These methods will be explained with the help of examples.

Art. 39. Show that the general equation of a straight line contains four arbitrary constants.

we can say that any equations.

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

represent a straight line passing through the point (x_1, y_1, z_1) and having direction ratios proportional to l, m, n . The equations of a straight line may be written as

$$\frac{x - x_1}{l} = \frac{y - y_1}{m}, \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

The line (1) meets the plane $x - y + z = 5$

Putting the values of x, y, z in (2)

$$\therefore 3r + 2 - 4r + 1 + 12r + 2 = 5 \text{ or, } r = 0$$

The co-ordinates of the point are $(2, -1, 2)$

$$\text{The required distance} = \sqrt{(2+1)^2 + (-1+5)^2 + (2+10)^2} = 13$$

Ex 12. (a) Find the distance of the pt. $(-1, 2, 5)$ from the st. line through the pt. $(3, 4, 6)$ whose direction cosines are propotional to $(2, -3, 6)$ [D. U. 1982, 82]

Actual directions are $\frac{2}{\sqrt{9+16+36}} = \frac{2}{7}, \frac{-3}{7}, \frac{6}{7}$. The distance,

$$\delta^2 = [(3 - (-1))^2 + (4 - 2)^2 + (5 - 5)^2] - [(3 - (-1)) \times \frac{2}{7} + (4 - 2) \times \frac{-3}{7} + (5 - 5) \times \frac{6}{7}]^2$$

$$= 16 + 4 - \{(8 - 6)/7\}^2 = 20 - 4/49 = 976/49$$

$$\therefore \delta = 4\sqrt{61}/7. \text{ By Art. 15.}$$

Ex 13. Find the shortest distance between the lines

$$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}, \quad \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$$

Find also equation and the points in which it meets the given lines

[C. U. 1986; C. U. (Hons.) 1987; D. U. 1970]

$$\text{Let } \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = r_1 \text{ or, } x = 3r_1 + 3, y = 8 - r_1, z = 3 + r_1$$

$$\text{and } \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} = r_2 \text{ or, } x = -3r_2 - 3, y = 2r_2 - 7, z = 4r_2 + 6$$

Let $L(3r_1 + 3, 8 - r_1, 3 + r_1)$; $M(-3r_2 - 3, 2r_2 - 7, 4r_2 + 6)$ by any two points L, M on the given lines

The direction ratios of the LM are

$$3r_1 + 3r_2 + 6, -r_1 - 2r_2 + 15, r_1 - 4r_2 - 3$$

If the line is perpendicular to both the given lines, then we have

$$3(3r_1 + 3r_2 + 6) - 1(-r_1 - 2r_2 + 15) + (r_1 - 4r_2 - 3) = 0$$

$$\text{and } -3(3r_1 + 3r_2 + 6) + 2(-r_1 - 2r_2 + 15) + 4(r_1 - 4r_2 - 3) = 0$$

Simplifying, we have $11r_1 + 7r_2 = 0$ and $7r_1 + 29r_2 = 0$

These two given $r_1 = r_2 = 0$ and hence the point, L and M are $(3, 8, 3)$ and $(-3, -7, 6)$.
equation of the line LM which is the S. D. is

$$\frac{x-3}{3+3} = \frac{y-8}{8+7} = \frac{z-3}{3-6} \text{ or, } \frac{x-3}{-2} = \frac{y-8}{-5} = \frac{z-3}{1}$$

The length of S. D. between L and M

$$= \sqrt{(3+3)^2 + (8+7)^2 + (3-6)^2} = \sqrt{270} = 3\sqrt{30}$$

Ex. 14. Find the magnitude and the equations of the line of shortest distance between

the lines

$$\frac{x-8}{3} = \frac{y+9}{-16} = \frac{z-10}{7} \dots \dots (1); \quad \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

Let l, m, n be the direction cosines of the lines L, M of shortest distance. As it is perpendicular two lines we have

$$3l - 16m + 7n = 0 \text{ and } 3l + 8m - 5n = 0$$

$$\therefore l/24 = m/36 = n/72 \text{ or } l/2 = m/3 = n/6 = 1/7$$

$$\text{Hence } l = 2/7, m = 3/7, n = 6/7$$

and (15, 29, 5) on

The magnitude of the shortest distance is the projection of the joint of the pts. (8, -9, 10),

$$ML \text{ and is } = (15-8)2/7 + (29+9)3/7 + (5-10)6/7 = 10 \quad \text{or} \quad S.D. = \sqrt{(x_2-x_1)^2 + m^2(y_2-y_1)^2 + n^2(z_2-z_1)^2}$$

Again the equation of the plane containing the first of the two given lines and the line of shortest distance is

$$\begin{vmatrix} x-8 & y+9 & z-10 \\ 3 & -16 & 7 \\ 2 & 3 & 6 \end{vmatrix} = 0 \quad \text{Or, } 117x + 4y - 41z - 490 = 0$$

Also the equation of the plane containing the second line and the S. D. is

$$\begin{vmatrix} x-15 & y-29 & z-5 \\ 3 & 8 & -5 \\ 2 & 3 & 6 \end{vmatrix} = 0 \quad \text{Or, } 9x - 4y - z - 14 = 0$$

The equation of the shortest distance is

$$117x + 4y - 41z - 490 = 0 = 9x - 4y - z - 14$$

Ex. 15. Through the pt. A (-1, 1, 2) a line is drawn parallel to the line of intersection of the planes $x - 2y + z = 3$ and $x + 6y - 5z = 0$ This line cuts the plane $x - 3y + 2z = 2$ in B. Find the equation of the line AB and the co-ordinates. [R. U. 1962]

The direction ratios of the line

$$x - 2y + z - 3 = 0 = x + 6y - 5z \dots \dots \dots (2)$$

are from

$$\therefore \begin{vmatrix} 1 & -2 & 1 \\ 1 & 6 & -5 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 6 & -5 \end{vmatrix} = -5 - 6 = -11 \text{ or } 11$$

The direction ratios of the line parallel to (1) are 2, 3, 4

The equation of the line through A (-1, 1, 2)

$$\text{are } \frac{x+1}{2} = \frac{y-1}{3} = \frac{z-2}{4} = r \text{ (say)} \dots \dots \dots (2)$$

$$\text{If (2) meets the plane } x - 3y + 2z - 2 = 0 \dots \dots \dots (3)$$

$$\text{then } (2r-1) - 3(3r+1) + 2(2+4r) - 2 = 0 \text{ or, } r = 2.$$